

Stability and Integrity of Aerospace Structures (SIAS)
Laboratory Session 1 Report

Structural Optimization of the Boeing 767-300 Wing

Students: Nuño Gallo Ochoa (100521973), Imad El Kadi (100521907), Ian Gallo Ochoa (100521932), Alfonso Martínez Magaz (100521999), Alexander Dost Belda (100522046)

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1 Introduction

The present report covers a comprehensive structural analysis and optimization of a conventional airliner wing. Specifically, a study of the Boeing 767-300 will be conducted, encompassing a preliminary identification of the wing and wing box main dimensions and aerodynamic and structural loads. This is followed by an idealization process prior to the final optimization, which is based on the positioning of the three longitudinal spars comprising the wing box. Finally, a Finite Element Method (FEM) analysis will be performed on the final, lightest structure for validation purposes.



Figure 1: Delta Air Lines' Boeing 767-300ER.

2 Preliminary Analysis

2.0.1 Identification of the Wing Configuration

First, the planform geometry and overall dimensions of the Boeing 767-300 were obtained from the official Airplane Characteristics for Airport Planning document provided by the manufacturer.

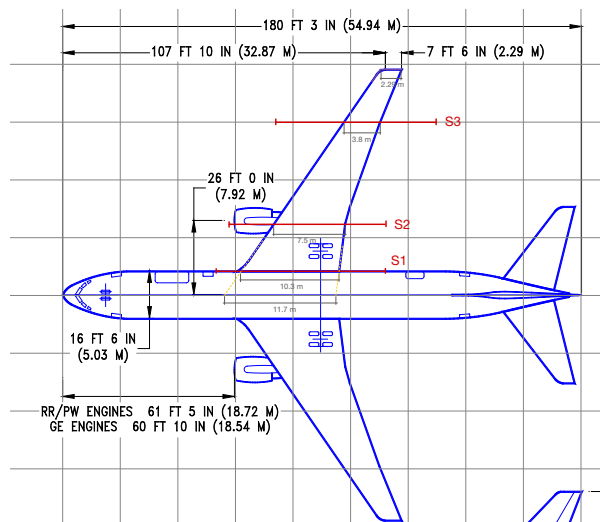


Figure 2: Wing planform

As can be observed in **Figure 2**, the wing planform exhibits a trapezoidal layout featuring a noticeable discontinuity in the trailing-edge sweep angle at section 2, commonly referred to as the kink. This geometric feature divides the wing into an inboard portion, extending from the fuselage junction at

Section	Variable	Chord (m)
Center line	c_{CL}	11.7
Wing root (S_1)	c_R	10.3
Kink (S_2)	c_K	7.5
75% semi-span (S_3)	$c_{0.75}$	3.8
Wing tip	c_T	2.29

Table 1: Wing Chord Lengths by Station

Parameter	Variable	Span (m)
Wingspan	b	47.57
Kink position	b_K	15.84
Root position	y_R	5.03

Table 2: Wing Spanwise Parameters

section 1 up to the kink, and an outboard segment that stretches from the kink towards the wing tip. Furthermore, an additional section 3 is defined at 75% of the semi-span; its geometric properties, along with those of the other target sections, are detailed in **Table 3**.

Target Section	y_i (m)	c_i (m)	Airfoil
Wing root (S_1)	5.03	10.30	NACA 0014
Kink (S_2)	15.84	7.50	NACA 0012
75% semi-span (S_3)	17.84	3.80	NACA 0009

Table 3: Target cross-sections for the structural sizing process.

The inboard region has a substantial chord and pronounced taper to meet structural requirements, such as integrating the landing gear. In the outboard region, the wing adopts a higher aspect ratio with a more gradual taper, intended to enhance aerodynamic efficiency. Moreover, both regions maintain a constant leading-edge sweep angle along the span.

Conversely, the internal wingbox maintains a uniform trapezoidal shape along the entire span. Its cross-sectional area decreases progressively towards the tip due to the external taper, yet the relative positions of the structural components stay proportionate to the local chord, both inboard and outboard of the kink.

2.1 Determination of the lift distribution

Prior to performing aerodynamic calculations, it is convenient to define an equivalent trapezoidal wing to achieve a linear chord distribution and simplify the aerodynamic model. In this way, the additional surface area corresponding to the kink is redistributed along the span by sweeping the straight trailing edge backwards, ensuring that the total planform area remains unchanged in the equivalent geometry. To that effect, the surface of the trapezoidal regions inboard and outboard of the kink and the area covered by the fuselage must be calculated, allowing to determine the wing exposed area:

$$\begin{cases} S_{in} = 0.5 \cdot (c_K + c_{CL}) \cdot b_K = 144 \text{ m}^2 \\ S_{out} = 0.5 \cdot (c_K + c_T) \cdot (b - b_K) = 159.43 \text{ m}^2 \\ S_{fus} = 0.5 \cdot (c_R + c_{CL}) \cdot y_R = 55.33 \text{ m}^2 \end{cases} \implies S_E = (S_{in} + S_{out} - S_{fus}) = 248.1 \text{ m}^2$$

Additionally, several key geometric properties are defined for subsequent calculations, including the leading-edge sweep angle (Λ_{LE}), the mean aerodynamic chord ($\bar{c}$ or MAC) and the aspect ratio (AR_e). The wing's geometric twist, introduced to delay tip stall and maintain aileron effectiveness (washout), follows a linear spanwise distribution resulting in a 2° nose-down angle at the tip.

$$\bar{c} = \frac{S_E}{b} = 5.22 \text{ m} \quad \Lambda_{LE} = 34.5^\circ \quad AR_e = \frac{b^2}{S_E} = 9.12$$

Subsequently, the geometric characteristics of the equivalent wing planform may be calculated, including the chord distribution at the root and centerline (c_{Re} and c_{CLE} , respectively), the wing taper

ratio (λ_e) and the quarter-chord sweep angle ($\Lambda_{c/4}^e$). Note that this sweep angle is slightly lower than that found at the leading edge due to taper, and aligns closely with online aviation literature suggesting that $\Lambda_{c/4}^e$ is about 31.5° in this aircraft.

$$c_{Re} = \frac{2 \cdot S_E}{b - y_R} - c_T = 9.37 \text{ m}$$

$$c_{CLe} = \frac{b \cdot c_{Re} - y_R \cdot c_T}{b - y_R} = 10.21 \text{ m}$$

$$\lambda_e = \frac{c_T}{c_{CLe}} = 0.224$$

$$\Lambda_{25}^e = \arctan \left[\tan \Lambda_{LE} - 0.5 \cdot \frac{c_{CLe}}{b} \cdot (1 - \lambda_e) \right] = 31.13^\circ$$

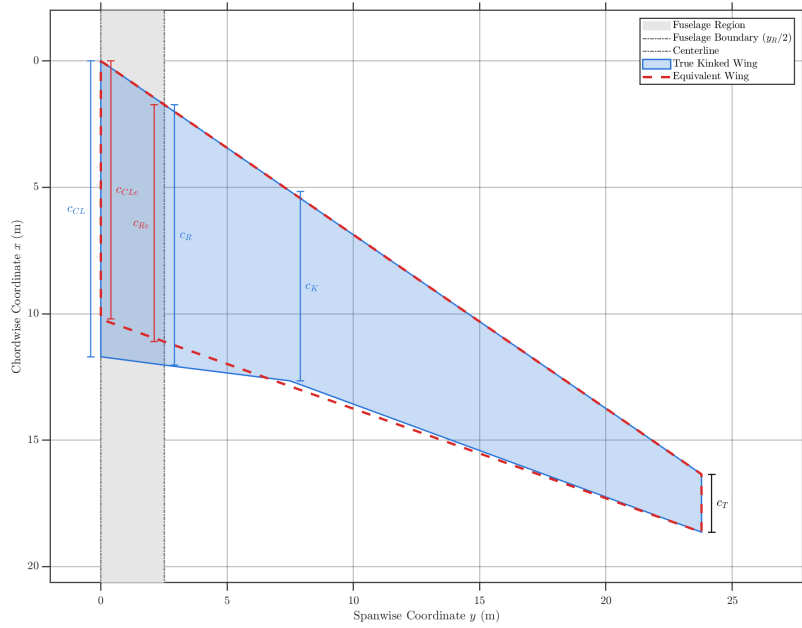


Figure 3: Real and equivalent Boeing 767-300 wing planform.

With the geometric properties of the reference wing established, the aerodynamic model used to determine the lift distribution can be introduced. Specifically, this analysis will employ the approximate method for subsonic wings proposed by Franklin W. Diederich in NACA Technical Note 2751. To ensure a rigorous assessment, the limiting operational condition at cruise ($M = 0.8$) is evaluated, assuming that the aircraft is flying at its maximum takeoff weight (MTOW) of 158,758 kg at its flight ceiling of 43,000 ft, where both density and temperature are at a minimum within the flight envelope, meaning that the lift coefficient will take its maximum value. Moreover, a load factor of $n_z = 2.5$ will be applied to account for pull-up and turning maneuvers. Under these conditions, the global lift coefficient of the wing (C_L) is calculated by equating the aerodynamic lift to the operating weight, multiplied by the corresponding load factor, as indicated below. The true aerodynamic velocity (V_c) is derived using the local speed of sound as $V_c = M\sqrt{\gamma R_g T}$, and the air density (ρ_c) is calculated according to the International Standard Atmosphere (ISA) model using the `atmosisa` function in MATLAB.

$$C_L = \frac{2 \cdot n_z \cdot W_{\max}}{S_E \cdot \rho_c \cdot V_c^2} = 2.157$$

Firstly, this method is based on the Prandtl-Glauert compressibility correction to adapt well-established aerodynamic results from incompressible potential flow theory. Specifically, it modifies the two-dimensional airfoil lift curve slope (c_{l_α} , which equals 2π per radian at $M \approx 0$), as well as the three-dimensional wing lift curve slope (C_{L_α}). By appropriately defining the compressibility factor (β) and the compressibility-corrected sweep angle (Λ_β) as:

$$\beta = \sqrt{1 - M^2} = 0.6$$

$$\Lambda_\beta = \arctan \left(\frac{\tan \Lambda_{c/4}^e}{\beta} \right) = 45.5^\circ$$

the corresponding expressions for c_{l_α} and the C_{L_α} yield:

$$c_{l_\alpha} = \frac{2 \cdot \pi}{\sqrt{1 - (M \cdot \cos \Lambda_{25}^e)^2}} = 8.60 \quad C_{L_\alpha} = \frac{2 \cdot \pi \cdot AR_e}{2 + \sqrt{\left(\frac{2 \cdot \pi \cdot AR_e}{\cos \Lambda_\beta \cdot c_{l_\alpha}} \right)^2 + 4}} = 8.09$$

Diederich's method is a semi-empirical technique that determines spanwise lift distribution through the principle of superposition, as it calculates total lift by summing two distinct components: the basic lift distribution (L_B), which arises from the wing's geometric twist, and the additional lift distribution (L_A), which is driven by the wing's planform geometry and angle of attack, assuming a completely untwisted wing. The resulting expression for the spanwise lift distribution, normalized by the MAC and evaluated along the dimensionless span $\eta = 2y/b$, is defined as:

$$\frac{C_l(\eta) \cdot c(\eta)}{\bar{c}} = C_{L_w} \cdot L_A(\eta) + \epsilon_T \cdot C_{L_\alpha} \cdot L_B(\eta)$$

where C_{L_w} represents the wing's lift coefficient, which can be approximated as $C_{L_w} \approx 1.05 \cdot C_L$. Recalling that linear functions were considered for the chord and twist distributions:

$$\begin{cases} c(\eta) = c_{CLe} \left[1 - (1 - \lambda_e) \cdot \eta \right] \\ \epsilon(\eta) = \epsilon_T \cdot \eta \end{cases}$$

the expressions for the additional and basic lift distributions may be retrieved:

$$\begin{cases} L_A(\eta) = C_1(F) \cdot \frac{c(\eta)}{\bar{c}} + C_2(F) \cdot \frac{4}{\pi} \cdot \sqrt{1 - \eta^2} + C_3(F) \cdot f(\Lambda_\beta, \eta) \\ L_B(\eta) = L_A(\eta) \cdot C_4(F) \cdot \left[\frac{\epsilon(\eta)}{\epsilon_T} - \alpha_{0_1} \right] \end{cases} \quad \text{with} \quad \begin{cases} \alpha_{0_1} = \int_0^1 \frac{\epsilon(\eta)}{\epsilon_T} \cdot L_A(\eta) \cdot d\eta \\ F = \frac{2 \cdot \pi \cdot AR}{c_{l_\alpha} \cdot \cos(\Lambda_{25}^e)} \end{cases}$$

Both contributions depend on the planform similitude parameter (F) through four coefficients. These constants can be determined from the graph in **Figure 4** by evaluating the analytic value of $F \approx 7.7653$.

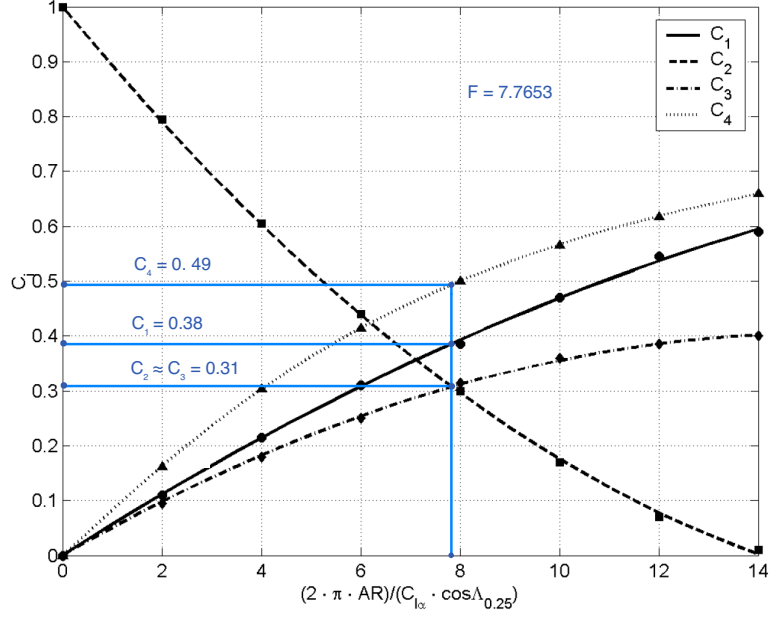


Figure 4: Identification of lift distribution coefficients in Diederich's method.

Similarly, the sweep correction function f depends on both the dimensionless spanwise coordinate η and the compressibility-corrected sweep angle, which is approximately 45° . The corresponding curve for $f(45^\circ, \eta)$ is presented in **Figure 5**. To compute the lift distribution, this curve must be digitized; for example, an online tool developed by the Department of Automotive and Aeronautical Engineering at HAW Hamburg provides a sixth-order polynomial approximation of this curve, which is displayed in **Figure 6**.

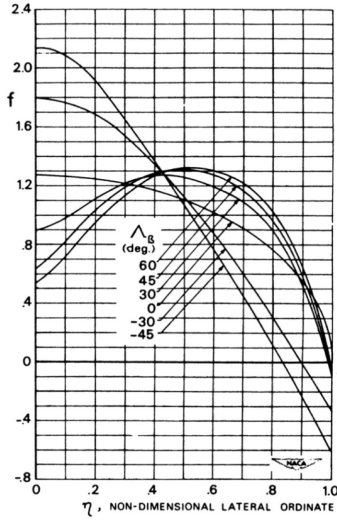


Figure 5: Sweep correction function from Torenbeek (1988).

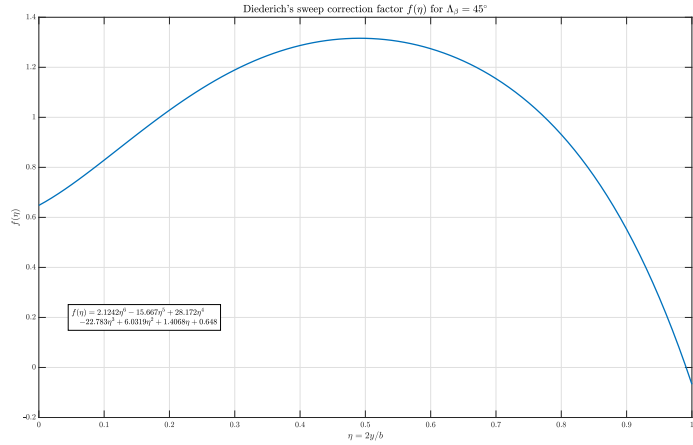


Figure 6: Sweep correction function $f(45^\circ, \eta)$ and its 6th-order polynomial approximation.

Finally, the basic lift distribution depends on the geometric twist of the wing both explicitly and through the dimensionless constant α_{01} . Because the basic lift distribution is defined such that the local aerodynamic forces generate no net lift, this constant α_{01} mathematically represents the shift in the zero-lift angle of attack resulting from the wing's twist. With all the aforementioned parameters established, the final lift distribution can be obtained and used to identify the local structural loads.

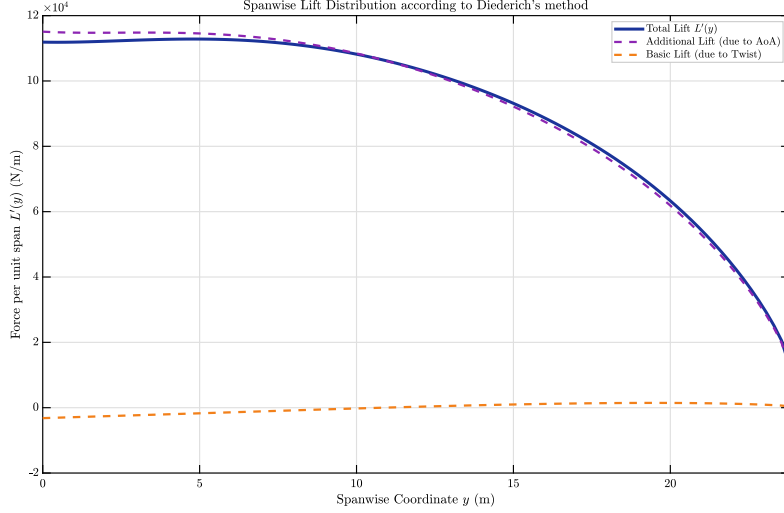


Figure 7: Spanwise lift distribution computed via Diederich's method at $M = 0.8$, $h = 43,000$ ft, and a load factor of $n_z = 2.5$.

2.2 Loads calculation

To analyze the structural response at the three predefined target sections, it is first necessary to determine the internal loads, namely, the shear force (Q_y), bending moment (M_x), and torsional moment (T) acting on them. This requires a precise definition of the external load spectrum applied to the semi-wing structure. These loads are modeled both as continuous distributions, inherently dependent on the local wing geometry, and as discrete point forces associated with the inertial effects of integrated components. The corresponding magnitudes and analytical expressions defining these external loads are summarized in **Table 4**.

Load	Type	Analytical Expression	Application Region
Aerodynamic Lift (L')	Distributed	Detailed in Section 2.1	$0 \leq z \leq b/2$
Wing Weight (q_{wing})	Distributed	$M_{wing} \cdot g \cdot n_z \left(\frac{c(z)}{S_E} \right)$	$0 \leq z \leq b/2$
Fuel Weight (q_{fuel})	Distributed	$\rho_{fuel} \cdot g \cdot n_z \cdot c(z)^2 \cdot NDSA(z)$	$0 \leq z \leq 0.8 \frac{b}{2}$
Engine Weight (W_{engine})	Point Load	$M_{engine} \cdot g \cdot n_z$	$z = z_{engine}$
Landing Gear (W_{gear})	Point Load	$M_{gear} \cdot g \cdot n_z$	$z = z_{gear}$

Table 4: Analytical expressions and application regions of the external loads applied to the semi-wing.

Note that the geometrical and mass properties of the wing, landing gear, engine, and fuel have been retrieved from the published B767-300 ACAP documentation and can be checked for reference in **Table 5**.

Component	Mass / Density	Spanwise Location (z)
Wing Structure	10258 kg	—
Engine	4410 kg	7.92 m
Landing Gear	3270 kg	4.65 m
Fuel (Jet A-1)	804 kg/m ³	—

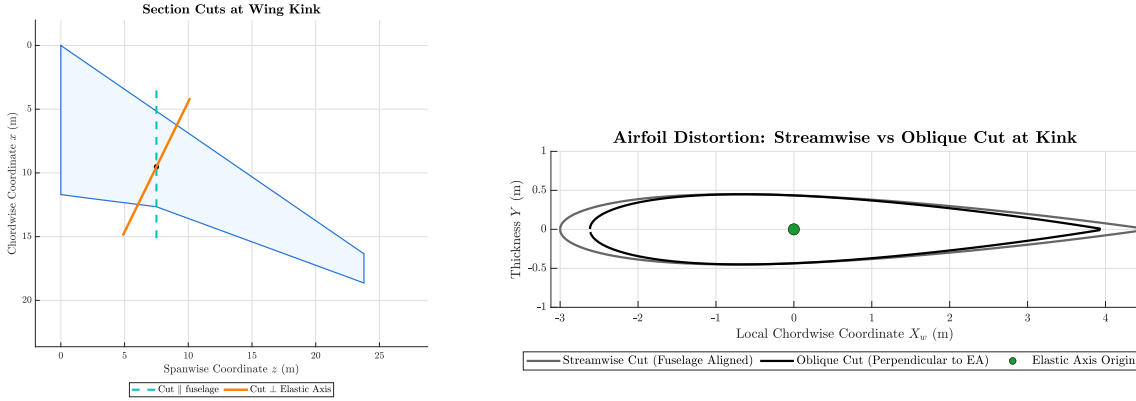
Table 5: Mass properties and locations of the main semi-wing components.

To model the fuel weight distribution, the internal volume available for fuel storage must be determined. This is defined through the Non Dimensional Structural Area (NDSA), previously introduced in the fuel configuration section. The NDSA represents the cross-sectional area of the wing box —bounded by the front and rear spars and the upper and lower skins—normalized by the chord. Thus, the dimensional area at a spanwise station z is obtained as $\text{NDSA}(z) \cdot c(z)^2$.

$$\text{NDSA} = \int_{\varepsilon_{fs}}^{\varepsilon_{rs}} \left[\left(\frac{y}{c} \right)_{up} - \left(\frac{y}{c} \right)_{low} \right] d\varepsilon$$

The application of the Navier–Bresse formulation for bending and torsion analysis requires the evaluation of structural cross-sections that are normal to the Elastic Axis (EA), which is defined as the geometric locus of the shear centers of all transverse sections. In a first order approximation, it will be assumed to be located upon the geometric center of the wingbox.

However, the three-dimensional wing geometry is generated by lofting aerodynamic profiles oriented parallel to the fuselage symmetry plane. As a result, any cutting plane normal to the swept EA is not aligned with these nominal airfoils. Since the structural cut intersects the lofted geometry obliquely, the resulting cross-section does not preserve the original thickness-to-chord ratio of a two-dimensional profile. Instead, the apparent thickness increases progressively from the leading edge to the trailing edge due to the projection induced by the sweep angle. This phenomenon is illustrated in **Figure 8**.



(a) Planform view of an aerodynamic section (parallel to free stream flow) and structural section normal to EA.

(b) Cross section profiles for aerodynamic and structural cuts.

Figure 8: Geometrical illustration for structural and aerodynamic applications.

To evaluate the internal structural response, the analysis is performed in a local coordinate system (x_w, y_w, z_w) aligned with the EA, instead of the global aircraft frame (x, y, z) . This transformation corresponds to a planar rotation about the vertical y -axis by the elastic sweep angle Λ_{EA} . It should be noted that Λ_{EA} is not a fixed planform parameter, but depends on the chordwise positions of the front and rear spars, which ultimately define the EA trajectory. As a result, due to wing taper, the EA generally exhibits a larger sweep angle than the leading edge. The inherent rotation of the geometry is shown in **Figure 9**.

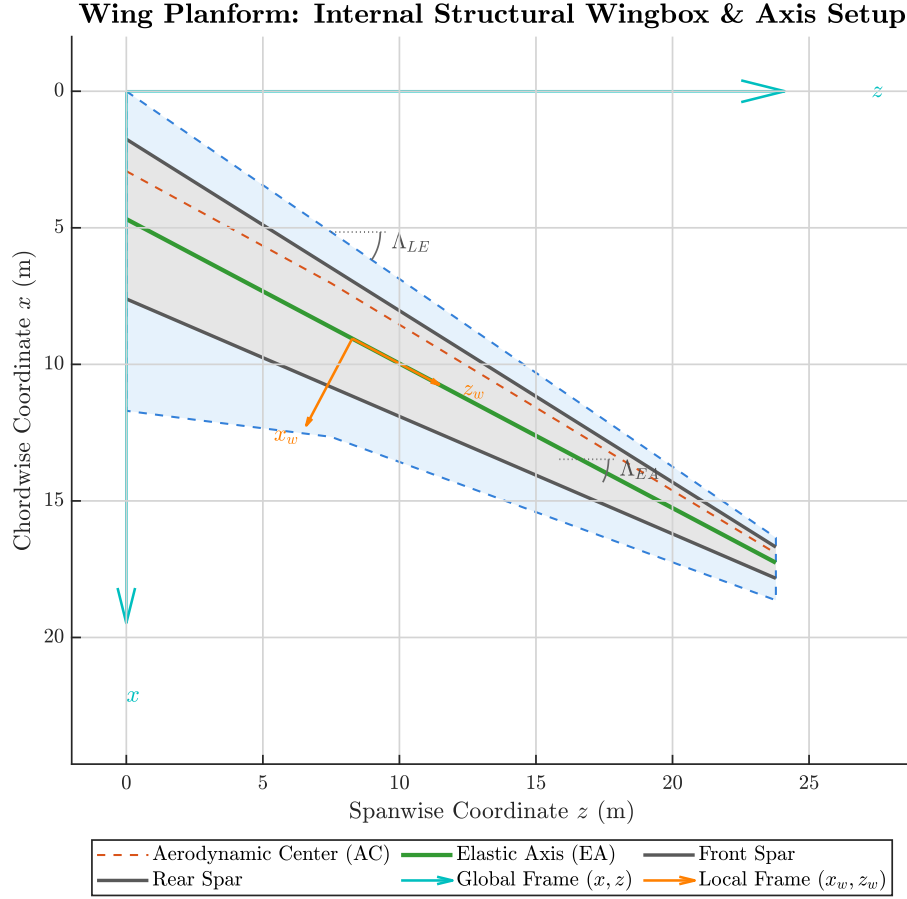


Figure 9: Aircraft and wing reference frames alongside force locus illustration and wingbox/elastic axis layout.

This geometric complexity makes the definition of external loads as continuous two-dimensional pressure fields over the wing surface impractical. To maintain analytical tractability while preserving structural consistency, the loads are simplified and represented as spanwise linear distributions. In this approach, the problem is reduced to one dimension, with loads expressed per unit span as functions of the spanwise coordinate z or z_w .

To this effect, inertial forces are assumed to be applied directly along the elastic axis, while the aerodynamic lift acts at the local aerodynamic center. Consequently, the inertial loads do not induce any twisting moment on the structure. The only torsional moment experienced by the wingbox is generated by the lift force, which is offset from the EA towards the leading edge, thereby creating a spanwise torsional moment arm.

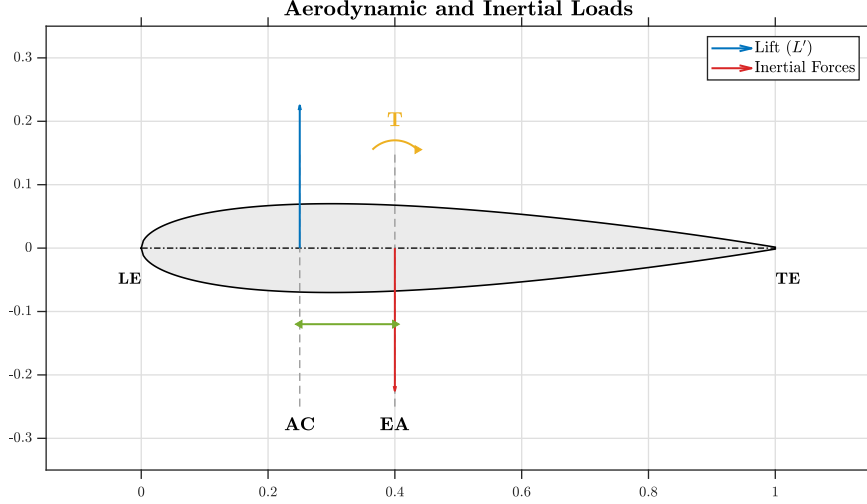


Figure 10: Force locus illustration at cross section (in this draft the wing tip would be located behind the given section).

Having established the precise geometric parameters and the complete spectrum of external loads acting upon the semi-wing structure, the analysis now proceeds to the evaluation of its internal structural response.

2.2.1 Shear force Q_y

To streamline the structural analysis, the continuous force distributions are integrated along the span. By modeling the semi-wing as a cantilever beam, the integration is performed from the free end at the tip of the wing, where the internal shear forces and bending moments are zero. This provides a well-defined starting point to accumulate the external loads inboard towards the wing root, up to each target section z_t .

The vertical shear force, $Q_y(z)$, at a specific structural target section is obtained by integrating the distributed vertical loads and summing the discrete point forces acting outboard of that station. Since the cross-section is taken perpendicularly to the swept EA, the integration limit for the aerodynamic lift is adjusted to $z_{L,lim}$ to precisely capture the reduced aerodynamic area outboard of the structural cut. Conversely, the inertial masses are integrated purely along the span from z_t to the tip ($b/2$). Such an offset in the integration limits can be observed in **Figure 8**, presented earlier for reference.

To determine the precise aerodynamic integration limit, $z_{L,lim}$, the intersection between the oblique structural cutting plane (perpendicular to the EA) and the 25% aerodynamic chord line, $x_L(z)$, must be calculated. For a given target station z_t , the perpendicular cutting line is defined using the normal slope $m_{normal} = -1/m_{EA}$:

$$x_{cut}(z) = x_{EA}(z_t) + m_{normal} \cdot (z - z_t)$$

By equating this structural cut with the local lift line equation ($x_L(z) = x_{L,0} + m_L \cdot z$), the geometric intersection point is analytically isolated as:

$$z_{L,lim} = \frac{x_{EA}(z_t) - x_{L,0} - m_{normal} \cdot z_t}{m_L - m_{normal}}$$

Due to the wing planform kink, this analytical intersection is evaluated piecewise for both the inboard (m_{L1}) and outboard (m_{L2}) regions to extract the geometrically valid root within the physical span. Moreover, this geometric system is also evaluated piecewise for both the inner and outer wing bays to extract the valid spanwise root within the $[0, b/2]$ interval.

At last, the analytical expression implemented for the total shear force becomes:

$$Q_y(z_t) = \int_{z_{L,lim}(z_t)}^{b/2} L'(z) dz - \int_{z_t}^{b/2} [q_{wing}(z) + q_{fuel}(z)] dz - \sum_{z_i \geq z_t} P_i$$

Where P_i represents the point loads due to the engine and the main landing gear located at spanwise stations z_i outboard of the evaluated section. Note that unlike the bending and torsional moments, the vertical shear force does not require a trigonometric transformation to the local coordinate system, as the vertical axis (y) is shared between the global aircraft reference frame and the local structural cross-section.

2.2.2 Bending moment M_{x_w}

Following the determination of the shear forces, the global bending moment, $M_{x_w}(z)$, at the evaluated cross-section is derived. This moment is computed by accumulating the bending effects generated by all vertical external forces acting outboard of the target station. Mathematically, it is expressed as the integral of the distributed loads multiplied by their respective spanwise lever arms, $(z - z_t)$, relative to the cutting plane:

$$M_x(z_t) = \int_{z_{L,lim}}^{b/2} L'(z) \cdot (z - z_t) dz - \int_{z_t}^{b/2} [q_{wing}(z) + q_{fuel}(z)] \cdot (z - z_t) dz - \sum_{z_i \geq z_t} P_i \cdot (z_i - z_t)$$

It is important to note that this equation yields the bending moment about the global chordwise axis (x). Because the structural wing box is swept, this global moment, alongside the global torsional moment, will subsequently require a trigonometric transformation to be projected onto the local structural axes of the cross-section.

2.2.3 Torsional moment T

Concurrently with the global bending moment, the global torsional moment, $T(z_t)$, must be integrated prior to any axis transformation. Assuming the inertial masses (wing structural weight and fuel) act directly through the local EA, they do not induce primary torsion. Thus, the global torsional moment is driven entirely by the aerodynamic lift. It is computed by integrating the lift distribution multiplied by the local chordwise torsion arm, defined as the offset between the EA and the 25% aerodynamic center line ($\Delta x(z) = x_{EA}(z) - x_L(z)$):

$$T(z_t) = \int_{z_{L,lim}}^{b/2} L'(z) \cdot [x_{EA}(z) - x_L(z)] dz$$

2.2.4 Numerical results and discussion

Finally, the global moments must be projected onto the local structural coordinate system (x_w, z_w), which is aligned with the swept EA. Using the elastic sweep angle Λ_{EA} , this trigonometric transformation resolves the true local bending (M_{x_w}) and torsional (T) moments applied to the cross-section:

$$\begin{cases} M_{x_w}(z_t) = M_x(z_t) \cos(\Lambda_{EA}) - M_z(z_t) \sin(\Lambda_{EA}) \\ T(z_t) = M_x(z_t) \sin(\Lambda_{EA}) + M_z(z_t) \cos(\Lambda_{EA}) \end{cases}$$

The obtained internal loads for the three target sections at the fuselage intersection, kink, and three quarters of semi-span are summarized in the table below.

Target Section	z (m)	Q_y (kN)	M_{x_w} (kN·m)	T (kN·m)
Fuselage	2.515	+748.89	+8610.16	+7073.70
Kink	7.500	+718.46	+5405.27	+4342.55
75% Span	17.839	+251.05	+535.99	+473.48

Table 6: Final Local Structural Loads at Target Sections.

To support the numerical results, the stress characterist plots are shown in the figure below. The obtained trends are consistent with the expected behavior of a cantilever beam, representing successive section cuts along the wing span towards the fixed root.

Due to the high aerodynamic lift, a significant **downward reaction force develops at the root constraint, reaching 698.7 kN**. As a result, the internal shear force at any given section cut including the root must be positive in the vertical direction to balance this large negative reaction.

Under this sign convention, concentrated downward loads such as the engine weight and landing gear, act in the same direction as the root reaction, leading to abrupt increases in the shear force diagram in order to satisfy global equilibrium. As expected, the maximum internal loads occur close to the wing root, making it the most critical structural region. Consequently, the inboard sections of the wing require more demanding structural sizing, typically involving thicker skin panels and larger stringer cross-sections.

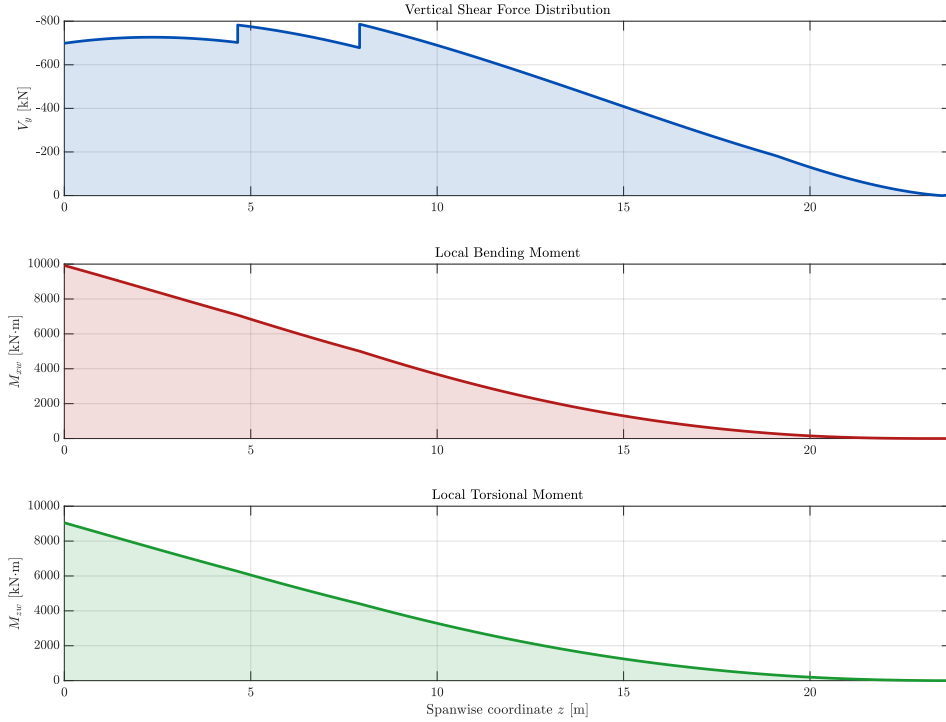


Figure 11: Stress characteristic plots along wing structure.

Note: While the prior numerical results strictly correspond to the 15%, 53%, and 65% chord positions of the spars, and as such are not universal for the wing structure, they are indeed representative of the overall bending response of a cantilevered beam. Crucially, these results serve to illustrate the precise order of magnitude of the structural forces addressed in this preliminary design stage.

The structural layout and internal load evaluation are inextricably linked. The development and distribution of loads within the wing are directly controlled by the arrangement of spars. Changing their chordwise position impacts the fuel tank geometry, which at the same time affects mass distribution

and internal forces, the EA sweep and the local reference frame, the distance between the Aerodynamic Center and the EA and, consequently, the torsional moment T. The internal loads must therefore be regularly updated each time the layout changes.

3 Sizing Process and Structural Idealization

With the spanwise distribution of internal loads (M_{xw} , Q_y , and T) already determined, the structural sizing of the wingbox is performed in this section using the fully-stressed design paradigm. To translate these internal loads into local stresses, a prior idealization of the target cross-sections (S1, S2 and S3) airfoils must be performed. Furthermore, the wingbox will be constructed using Al2024-T3, and the properties of this alloy will be obtained from the corresponding section of the U.S. Department of Defense Handbook MIL-HDBK-5H, as presented in the table below.

Table 3.2.3.0(b). Design Mechanical and Physical Properties of 2024 Aluminum Alloy Sheet and Plate																				
Specification	AMS 4037 and AMS-QQ-A-250/4															AMS-QQ-A-250/4				
Form	Sheet					Plate										Sheet		Plate		
Temper	T3					T351										T361				
Thickness, in.	0.008-0.009	0.010-0.128	0.129-0.249		0.250-0.499		0.500-1.000		1.001-1.500		1.501-2.000		2.001-3.000		3.001-4.000		0.020-0.062	0.063-0.249	0.250-0.500	
Basis	S	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	S	S	S
Mechanical Properties:																				
F_u , ksi:																				
L	64	64	65	65	66	64	66	63	65	62	64	62	64	60	62	57	59	68	69	67
LT	63	63	64	64	65	64	66	63	65	62	64	62	64	60	62	57	59	67	68	66
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	52*	54*	49*	51*	...	...	...
F_u , ksi:																				
L	47	47	48	47	48	48	50	48	50	47	50	47	49	46	48	43	46	56	56	54
LT	42	42	43	42	43	42	44	42	44	42	44	42	44	42	44	41	43	50	51	49
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	38*	40*	38*	39*	...	...	...
F_u , ksi:																				
L	39	39	40	39	40	39	41	39	41	39	40	38	40	37	39	35	37	47	48	46
LT	45	45	46	45	46	45	47	45	47	44	46	44	46	43	45	41	43	53	54	52
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	46	48	44	47	...	...	...
F_u , ksi:																				
F_{br} , ksi:																				
(e/D = 1.5)	104	104	106	106	107	97	100	95	98	94	97	94	97	91	94	86	89	111	112	109
(e/D = 2.0)	129	129	131	131	133	119	122	117	120	115	119	115	119	111	115	106	109	137	139	135
F_{br} , ksi:																				
(e/D = 1.5)	73	73	75	73	75	72	76	72	76	72	76	72	76	72	76	70	74	82	84	81
(e/D = 2.0)	88	88	90	88	90	86	90	86	90	86	90	86	90	86	90	84	88	97	99	96
e, percent (S-basis):	10	c	...	c	...	12	...	8	...	7	...	6	...	4	...	4	...	8	9	9 ^d
LT	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	10.5	10.5	10.7
E , 10 ³ ksi	10.5										10.7									
E_s , 10 ³ ksi	10.7										10.9									
G , 10 ³ ksi	4.0										4.0									
μ	0.33										0.33									
Physical Properties:																				
ω , lb/in.	0.100																			
C, K, and α	See Figure 3.2.3.0																			

a Caution: This specific alloy, temper, and product form exhibits poor stress-corrosion cracking resistance in this grain direction. It corresponds to an SCC resistance rating of D, as indicated in Table 3.1.2.3.1(a).
b Bearing values are "dry pin" values per Section 1.4.7.1. See Table 3.1.2.1.1.
c See Table 3.2.3.0(c).
d 10% for 0.500 inch.

Figure 12: Material properties of 2xxx series Aluminium alloys, sourced from MIL-HDBK-5J.

Note: In case any of these properties depend on the thickness of the panels or the kind of processing, average values will be taken for simplicity.

3.1 Section Idealization

In this idealization model, the spar caps are replaced by concentrations of area denoted as spar-booms, which are assumed to carry all the axial stresses resulting from bending moments. On the other hand, the skin panels and spar webs are assumed to carry only shear stresses, due to their small thickness, and they are simply defined as straight connections between the spar-booms. Furthermore, an inter-boom is placed at the midpoint of every skin segment between the primary spar-booms to refine the stress distribution.

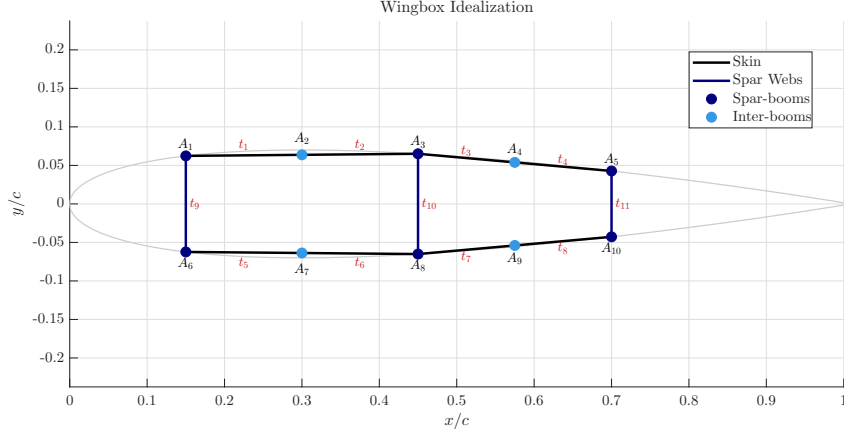


Figure 13: Example of idealized wingbox cross-section geometry.

As can be seen in **Figure 13**, for this Boeing 767-300 analysis, the wingbox is modeled as a two-cell structure composed by three spars webs (front, mid, and rear). This configuration results in a total of 10 boom areas (A_i), with five located on the upper surface and five on the lower surface, corresponding to the three spar cap locations and the two inter-boom positions for each surface. Regarding the skin and the webs, a total of 11 thicknesses (t_j) can be defined, dividing the panels into four upper skin segments, four lower skin segments, and three vertical spar webs. The final wingbox geometry of each section i is scaled according to the local chord and the specific thickness-to-chord ratio of their airfoils, as defined in **Table 3**.

The design variables to be employed in the following section for the optimization process in each section are precisely these boom areas and panel thicknesses, together with the combination of the vertical spar webs positioning.

3.2 Fully Stressed Paradigm

To ensure maximum structural efficiency, the sizing of these components is governed by the Fully Stressed Paradigm, whose main objective is to minimize the total weight of the wingbox structure by imposing that every element, both booms and skins, operates at its maximum allowable limit under the critical design load case.

In practice, to follow this paradigm, the cross-sectional areas A_i and thicknesses t_j must be iteratively adjusted for each of the three target sections by means of an optimization algorithm until the reserve factor (RF) for each element converges to unity, representing the fully stressed state. It is crucial to include a 30% safety factor into the criteria for the allowable σ_{max} and τ_{max} with respect to the yielding critical values, to keep a structural failure margin against uncertainties. Consequently, and taking into account that the yielding values for the wingbox material AL 2024-T3 are $\sigma_y = 345$ MPa and $\tau_y = 207$ MPa, the following value of the normal and shear stresses must be applied.

$$\begin{cases} RF_\sigma = \frac{\sigma_{max}}{\sigma_{applied}} = 1.0 \quad \rightarrow \quad \sigma_{max} = 0.7 \sigma_y = \sigma_{applied} = 241.5 \text{ MPa} & \text{(for booms)} \\ RF_\tau = \frac{\tau_{max}}{\tau_{applied}} = 1.0 \quad \rightarrow \quad \tau_{max} = 0.7 \tau_y = \tau_{applied} = 144.9 \text{ MPa} & \text{(for panels)} \end{cases}$$

The booms areas are recalculated to satisfy the normal stress limit imposed by the bending moment (M_x) and, simultaneously, the skin and web thicknesses are adjusted to withstand the shear flows resulting from the wing's torsion (T) and transverse shear forces (Q_y).

The cross-sectional inertial properties must be updated at every iteration to reflect the evolving value of boom areas. For the purpose of this sizing procedure, 3D tapering effects between sections are neglected, treating the local geometry of each target station as a representative 2D cross-section. Moreover, the optimization will be bound by specific technological constraints as strictly imposing minimum values

of $t_{min} = 1.0$ mm for panels and $A_{min} = 100 \text{ mm}^2$ for booms, while maximum geometric limits are established to prevent structural clashes between adjacent internal members. Finally, the entire sizing process is conducted under the $n_z = 2.5$ pull-up maneuver, serving as the critical representative load case for the structural integrity of the structure.

3.3 Sizing Optimization Algorithm

The methodology adopted to obtain the previously defined Fully Stressed Design configuration is based on two classical analytical approaches: the Navier–Bresse theory for beam bending and the structural formulation for multi-cell wing boxes developed by Megson. By combining these frameworks, the stress state of the idealized boom–skin cross-section can be explicitly expressed as a function of its geometric parameters. This analytical basis yields the governing equations used within a Least Squares optimization scheme, enabling the desired iterative adjustment of the structural variables (A_i and t_j) so that the structure converges to the minimum-weight design.

3.3.1 Normal Stresses in Booms (Navier-Bresse Formulation)

As established in the structural idealization, the discrete spar caps and stringers (idealized as booms) are assumed to carry all the normal forces resulting from wing bending. To calculate these forces, the classical Navier-Bresse formulation is applied. Based on the defined local structural coordinate system—where z_w is the spanwise axis, x_w is the oblique chordwise axis, and y_w is the vertical axis—the calculation of normal stresses is explicit and direct. The normal stress (σ_{z_w}) acting along the spanwise direction for any given boom i is formulated as:

$$\sigma_{z_w,i} = \frac{M_{x_w} (I_{y_w y_w} \cdot y_{w,i} - I_{x_w y_w} \cdot x_{w,i})}{I_{x_w x_w} I_{y_w y_w} - I_{x_w y_w}^2}$$

It is important to clarify the boundary assumptions governing this generalized bending formulation. While the analysis safely assumes that there are no external axial forces applied to the section ($N_{z_w} = 0$) and no bending moments acting about the vertical axis ($M_{y_w} = 0$), structural symmetry is not artificially enforced. Although the external aerodynamic airfoil may be symmetric, the optimal cross sectional areas of the discrete booms (A_i) might not be distributed symmetrically. Consequently, the cross-sectional product of inertia cannot be inherently assumed as null ($I_{x_w y_w} \neq 0$). By retaining the full coupled formulation, the mathematical system is granted additional degrees of freedom. This robust approach allows the non linear optimizer to freely explore asymmetric structural layouts, freely converging toward a symmetric material distribution only if it mathematically represents the true, unconstrained minimum weight configuration.

3.3.2 Shear Stresses in Skin Panels (Megson’s Formulation)

The structural skins and spar webs are responsible for resisting the vertical shear force (Q_y) and the local torsional moment (T). The idealized wingbox was configured as a two-cell closed cross-section (a forward cell, A_2 , and an aft cell, A_1 , separated by a central spar web). Due to this multi-cell topology, the internal shear flows are statically indeterminate.

To resolve this indeterminacy and compute the operational shear stress (τ) in each of the 11 panels, Megson’s analytical method for multi-cell structures is embedded within the sizing algorithm. The procedure is executed in three distinct mathematical steps:

Step 1: Basic Shear Flow (q_b)

In the first step, the closed cells are artificially "cut" to render the structure statically determinate. The basic open-section shear flow distribution (q_b) generated by the vertical shear force is calculated incrementally from boom to boom using the standard shear flow equation:

$$dq_b = - \left(\frac{Q_y \cdot I_{yy}}{I_{x_w x_w} I_{yy} - I_{x_w y}^2} \right) A_i \cdot y_i + \left(\frac{Q_y \cdot I_{x_w y}}{I_{x_w x_w} I_{yy} - I_{x_w y}^2} \right) A_i \cdot x_i$$

Note: The algorithm retains the full asymmetric formulation to guarantee robustness across all spanwise stations, inherently simplifying when the cross-section is symmetric.

Step 2: Constant Shear Flows (q_{s1}, q_{s2})

The second step is to restore the physical continuity of the closed cells superimposing an unknown constant shear flow (q_{s1} and q_{s2}) onto each respective cell. These values are found by solving a 2×2 system governed by the following two physical boundary conditions.

The first condition is kinematic compatibility, which implies that the rate of twist ($\frac{d\theta}{dz}$) must be uniform across both adjacent cells. This is enforced by equating the contour integrals of the shear deformation:

$$\frac{1}{2A_1} \oint_1 \frac{q_b + q_{s1}}{t} ds = \frac{1}{2A_2} \oint_2 \frac{q_b + q_{s2}}{t} ds$$

The second condition is to enforce static equilibrium so that the sum of the internal moments generated by all shear flows around any longitudinal axis must equal the externally applied torsional moment (T):

$$T = \sum M_{qb} + 2A_1 q_{s1} + 2A_2 q_{s2}$$

Step 3: Total Shear Stress (τ)

For the third and final step, by solving this coupled system, the algorithm retrieves q_{s1} and q_{s2} . The total operational shear flow in any panel j is the linear superposition of the basic and constant flows ($q_{total,j} = q_{b,j} + q_s$). Finally, the shear stress evaluated within the optimization loop is defined as:

$$\tau_j = \frac{|q_{total,j}|}{t_j}$$

It is crucial to emphasize that Megson's solution is inherently implicit. The cross-sectional properties (I_{xwxw}, I_{yy}) and the shear flow integrals depend directly on the design variables (A_i, t_j), which are simultaneously the unknowns being solved for. Consequently, the sizing system ceases to be linear. This inherent non-linearity strictly requires the use of alternative numerical approaches, such as the non-linear Least Squares optimization algorithm deployed in this study to iteratively resolve the structural coupling.

3.3.3 Iterative Formulation for Stress Residuals Convergence

As noted in the previous section, the sizing methodology consists of enforcing a target $RF = 1.0$ for all structural components. For the idealized wingbox, this leads to a system of $21 + 3$ equations: 21 stress limit conditions and the 3 definitions of the cross sectional area moments of inertia ($I_{xwxw}, I_{yw yw}, I_{xw yw}$). This corresponds to an equal number of unknowns given by the 10 boom areas (A_i), the 11 skin thicknesses (t_i), and the three inertia properties themselves. Although this suggests a determined system, the simplicity of the loading assumptions, specifically pure uniaxial bending driving the normal stresses, introduces geometric redundancies that make the system inherently incompatible, strictly requiring a non linear Least Squares optimization to iteratively resolve it.

In addition, the problem is inherently non linear, since the shear flows and stresses depend on the cross sectional properties, which are also unknowns. As a result, the system cannot be solved using a standard linear formulation and is instead addressed using a non linear least squares approach, where the stress residuals are minimized iteratively to obtain consistent values for the boom areas and skin thicknesses.

To mathematically execute this optimization, the non linear least squares algorithm seeks to minimize the following objective function, $J(\mathbf{A}, \mathbf{t})$, which aggregates the squared residuals of all RF_σ and RF_τ for all $N = 10$ booms and $M = 11$ skin panels:

$$J(\mathbf{A}, \mathbf{t}) = \sum_{i=1}^N (RF_{\sigma,i}(\mathbf{A}, \mathbf{t}) - 1)^2 + \sum_{j=1}^M (RF_{\tau,j}(\mathbf{A}, \mathbf{t}) - 1)^2$$

Where $\mathbf{A} = [A_1, A_2, \dots, A_N]^T$ is the design vector of boom areas, and $\mathbf{t} = [t_1, t_2, \dots, t_M]^T$ is the design vector of skin thicknesses.

By iteratively minimizing this error function, the solver accounts for the implicit geometric coupling and converges to the optimal cross sectional distributions. While this formulation drives the solution toward a minimum weight configuration, a pure least squares approach may converge to a local minimum where some components have $RF < 1.0$, indicating structural failure. To prevent this, the objective function includes an asymmetric penalty. Any residual associated with an undersized component ($RF < 1.0$) is amplified by a penalty factor before squaring. This ensures that the optimizer remains within the feasible design space and that the final vectors $\mathbf{A}$ and $\mathbf{t}$ satisfy the structural integrity constraints.

For reference, the whole structural solver and optimization algorithm are presented below.

```

1 function obj_residuals = compute_sizing_residuals(X, M_xw, T, Vy, x_booms, y_booms, ...
2           cell_area_1, cell_area_2, ...
3           sigma_yield, tau_yield)
4
5     % 1. Evaluate cross-section stress state using current design variables (A, t)
6     [RF_sigma, RF_tau, ~, ~, ~, ~] = evaluate_stress_state(X, M_xw, T, Vy, ...
7           x_booms, y_booms, ...
8           cell_area_1, cell_area_2,
9           sigma_yield, tau_yield);
10
11     % 2. Define residuals (FSD Target: Reserve Factor = 1.0)
12     residual_sigma = RF_sigma - 1.0;
13     residual_tau    = RF_tau - 1.0;
14
15     % 3. Asymmetric Penalty (Structural Safety Constraint)
16     % If RF < 1 (yield failure), the residual is heavily magnified
17     % to strictly force the optimizer out of the unsafe design space.
18     penalty_factor = 50;
19     residual_sigma(residual_sigma < 0) = residual_sigma(residual_sigma < 0) * penalty_factor;
20     residual_tau(residual_tau < 0)    = residual_tau(residual_tau < 0) * penalty_factor;
21
22     % 4. Assembly of the Objective Function array for lsqnonlin solver
23     obj_residuals = [residual_sigma, residual_tau];
24 end

```

```

1 function [RF_sigma, RF_tau, I_xx, centroid_y, centroid_x, q_total] = evaluate_stress_state(X, M_xw, T, Vy, x_booms, y_booms, ...
2           cell_area_1, cell_area_2, sigma_yield, tau_yield)
3
4     % Extract current design iteration variables
5     A_booms = X(1:10);
6     t_skins = X(11:21);
7
8     % --- CROSS-SECTIONAL PROPERTIES (Implicit Coupling) ---
9     centroid_x = sum(A_booms .* x_booms) / sum(A_booms);
10    centroid_y = sum(A_booms .* y_booms) / sum(A_booms);
11    x_dist = x_booms - centroid_x;
12    y_dist = y_booms - centroid_y;
13
14    I_xx = sum(A_booms .* (y_dist.^2));
15    I_yy = sum(A_booms .* (x_dist.^2));
16    I_xy = sum(A_booms .* (x_dist .* y_dist));
17    inertia_den = (I_xx * I_yy) - (I_xy^2);
18
19    % --- NAVIER-BRESSE: NORMAL STRESSES ---
20    sigma_normal = abs((M_xw * I_yy / inertia_den) .* y_dist - ...
21                    (M_xw * I_xy / inertia_den) .* x_dist);
22
23    % --- MEGSON STEP 1: BASIC SHEAR FLOWS (Open Section) ---
24    delta_qb = - (Vy * I_yy / inertia_den) .* (A_booms .* y_dist) + ...
25              (Vy * I_xy / inertia_den) .* (A_booms .* x_dist);
26
27    panel_nodes = [1 2; 2 3; 3 4; 4 5; 5 6; 6 7; 7 8; 8 9; 9 10; 10 1; 8 3];
28    panel_length = zeros(1,11);
29    for i = 1:11

```

```

30     panel_length(i) = sqrt((x_booms(panel_nodes(i,2))-x_booms(panel_nodes(i,1)))^2 + ...
31                          (y_booms(panel_nodes(i,2))-y_booms(panel_nodes(i,1)))^2);
32 end
33
34 q_basic = zeros(1,11);
35 % Open cell integration paths
36 q_basic(1) = 0;          q_basic(2) = q_basic(1) + delta_qb(2);
37 q_basic(3) = 0;          q_basic(4) = q_basic(3) + delta_qb(4);
38 q_basic(5) = q_basic(4) + delta_qb(5); q_basic(6) = q_basic(5) + delta_qb(6);
39 q_basic(7) = q_basic(6) + delta_qb(7);
40 q_basic(10) = -delta_qb(1); q_basic(9) = q_basic(10) - delta_qb(10);
41 q_basic(8) = q_basic(9) - delta_qb(9);
42 q_basic(11) = -q_basic(2) - delta_qb(3);
43
44 % --- MEGSON STEP 2: CONSTANT SHEAR FLOWS (Closed Cells) ---
45 ds_over_t = panel_length ./ t_skins;
46
47 % Kinematic compatibility integrals (ds/t)
48 delta_1 = ds_over_t(1) + ds_over_t(2) + ds_over_t(11) + ds_over_t(8) + ...
49           ds_over_t(9) + ds_over_t(10);
50 delta_2 = ds_over_t(3) + ds_over_t(4) + ds_over_t(5) + ds_over_t(6) + ...
51           ds_over_t(7) + ds_over_t(11);
52 delta_12 = ds_over_t(11);
53
54 int_qb_1 = q_basic(1)*ds_over_t(1) + q_basic(2)*ds_over_t(2) - ...
55           q_basic(11)*ds_over_t(11) + q_basic(8)*ds_over_t(8) + ...
56           q_basic(9)*ds_over_t(9) + q_basic(10)*ds_over_t(10);
57
58 int_qb_2 = q_basic(3)*ds_over_t(3) + q_basic(4)*ds_over_t(4) + ...
59           q_basic(5)*ds_over_t(5) + q_basic(6)*ds_over_t(6) + ...
60           q_basic(7)*ds_over_t(7) + q_basic(11)*ds_over_t(11);
61
62 % Static Equilibrium: Moment of basic shear flows
63 compute_moment = @(q, xi, yi, xf, yf) q * (xi*yf - xf*yi);
64 moment_qb = 0;
65 for i = 1:11
66     moment_qb = moment_qb + compute_moment(q_basic(i), ...
67       x_booms(panel_nodes(i,1)), y_booms(panel_nodes(i,1)), ...
68       x_booms(panel_nodes(i,2)), y_booms(panel_nodes(i,2)));
69 end
70
71 % Assemble and solve the 2x2 Megson System [C]{q_s} = {R}
72 C11 = delta_1/(2*cell_area_1) + delta_12/(2*cell_area_2);
73 C12 = -delta_12/(2*cell_area_1) - delta_2/(2*cell_area_2);
74 R1 = int_qb_2/(2*cell_area_2) - int_qb_1/(2*cell_area_1);
75
76 C21 = 2*cell_area_1;
77 C22 = 2*cell_area_2;
78 R2 = T - moment_qb;
79
80 q_constant = [C11, C12; C21, C22] \ [R1; R2];
81 q_s1 = q_constant(1);
82 q_s2 = q_constant(2);
83
84 % --- MEGSON STEP 3: TOTAL SHEAR FLOWS AND STRESSES ---
85 q_total = zeros(1,11);
86 q_total(1) = q_basic(1) + q_s1; q_total(2) = q_basic(2) + q_s1;
87 q_total(3) = q_basic(3) + q_s2; q_total(4) = q_basic(4) + q_s2;
88 q_total(5) = q_basic(5) + q_s2; q_total(6) = q_basic(6) + q_s2;
89 q_total(7) = q_basic(7) + q_s2; q_total(8) = q_basic(8) + q_s1;
90 q_total(9) = q_basic(9) + q_s1; q_total(10) = q_basic(10) + q_s1;
91 q_total(11) = q_basic(11) - q_s1 + q_s2;
92
93 tau_shear = abs(q_total) ./ t_skins;
94
95 % --- EVALUATE COMPONENT RESERVE FACTORS ---
96 RF_sigma = sigma_yield ./ sigma_normal;
97 RF_tau = tau_yield ./ tau_shear;
98 end

```

4 Design of Experiment (DoE)

4.1 Design Philosophy

To explore the structural efficiency of the internal wing layout, a parametric Design of Experiments was conducted. The reference aircraft, the Boeing 767, falls under the *DSPAR* case category, which focuses on optimizing the chordwise positions of the wingbox spars. Modifying these spar locations significantly alters the shape, internal volume, and torsional stiffness of the structural cells, directly impacting the overall weight required to withstand the aerodynamic and inertial loads.

In accordance with the project guidelines, the parametric study explores the following variations for the spar positions, expressed as a percentage of the local chord (%c):

- **Front Spar:** 15%, 20%, 25%
- **Mid Spar:** 47%, 50%, 53%
- **Rear Spar:** 65%, 70%, 75%

To systematically evaluate the structural impact of these geometrical changes, the sizing algorithm described in the previous section was fully automated. A computational loop was implemented to iterate through and solve the fully-stressed design problem for each of the $3 \times 3 \times 3 = 27$ possible configurations. By recording the resulting transversal area metrics for each setup, the algorithm successfully identifies the specific spar arrangement that yields the lightest possible wingbox structure.

To establish a reliable comparison of structural lightness among the evaluated configurations, a representative weight metric, denoted as m_w , was defined. This parameter serves as a proxy for the total structural mass by computing the cumulative cross-sectional material area required at the three critical target sections: the fuselage root, the aerodynamic kink, and the 75% span station.

Mathematically, m_w is calculated as the sum of the concentrated boom areas (A_j) and the distributed skin panel areas (derived from the thickness t_k and the length l_k of each panel) for each analyzed section i :

$$m_w = \sum_{i=1}^3 \left(\sum_{j=1}^{10} A_j + \sum_{k=1}^{11} t_k l_k \right)_i$$

where $i \in \{1, 2, 3\}$ corresponds to the root, kink, and 75% span sections, respectively. The resulting metrics for all iterations of the parametric study, identifying the geometrical setup that yields the minimum structural weight, are reported below.

Furthermore, the results table also reports the Fully Stressed Design Error (e_{FSD}) for each iteration. Due to the imposed geometric bounds and structural continuity requirements to resist torsion, achieving a perfect FSD state where all elements simultaneously reach a Reserve Factor (RF) of exactly 1.0 is mathematically impossible, rendering the ideal optimization an incompatible system. Nevertheless, monitoring this metric is essential to ensure that the solver does not produce large stress deviations and to guarantee that no specific geometric configuration induces a numerical or structural singularity. The e_{FSD} quantifies the average deviation of the active components from the ideal target, mathematically defined as:

$$e_{FSD} = \frac{1}{N} \sum_{i=1}^N |RF_i - 1|$$

Case	Front Spar (%)	Mid Spar (%)	Rear Spar (%)	m_w (m ²)	e_{FSD}
1	15	47	65	0.257024	0.2214
2	15	47	70	0.258863	0.2136
3	15	47	75	0.265523	0.2269
4	15	50	65	0.257148	0.2424
5	15	50	70	0.258510	0.2243
6	15	50	75	0.264298	0.2323
7	15	53	65	0.256179	0.2740
8	15	53	70	0.257933	0.2387
9	15	53	75	0.264342	0.2403
10	20	47	65	0.268004	0.2168
11	20	47	70	0.271037	0.2118
12	20	47	75	0.277702	0.2263
13	20	50	65	0.265377	0.2445
14	20	50	70	0.270082	0.2290
15	20	50	75	0.274105	0.2398
16	20	53	65	0.266727	0.2890
17	20	53	70	0.269359	0.2573
18	20	53	75	0.273023	0.2609
19	25	47	65	0.283841	0.2170
20	25	47	70	0.285942	0.2155
21	25	47	75	0.292783	0.2331
22	25	50	65	0.279076	0.2546
23	25	50	70	0.284762	0.2417
24	25	50	75	0.290944	0.2529
25	25	53	65	0.283266	0.3051
26	25	53	70	0.283189	0.2722
27	25	53	75	0.289065	0.2757

Table 7: Design of Experiments Results: Structural Weight Metric and FSD Error for all 27 configurations.

4.2 Optimal Design Discussion

As evidenced by the DoE results, the configuration yielding the minimum structural weight corresponds to Case 7, which positions the front, mid, and rear spars at 15%, 53%, and 65% of the local chord, respectively. This specific arrangement places the rear spar relatively close to the leading edge compared to the alternative evaluated setups (i.e., 70% or 75%). Consequently, the internal volume of the structural wingbox is reduced, which directly translates to a lower overall fuel mass capacity within the wing. Furthermore, restricting the wingbox boundary to the 65% chord station shifts the centroid of the fuel mass forward along the airfoil. This forward concentration of downward inertial weight more effectively opposes the upward aerodynamic lift acting near the quarter-chord. By providing this localized inertial relief, the coupled torsional and bending demands on the structure are significantly mitigated, allowing the optimizer to converge on a lighter design. The finalized cross-sectional areas for the booms, skin panel thicknesses, internal shear flows, and the corresponding Reserve Factors (RF) for each component of this optimal configuration are detailed below.

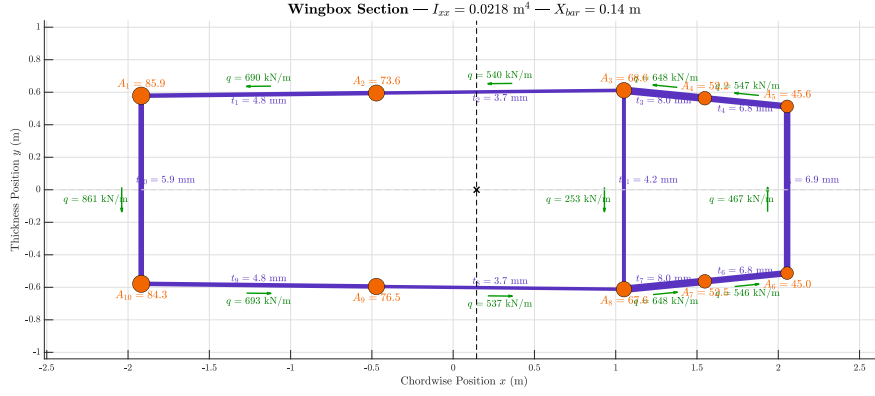


Figure 14: Fully stressed design cross-section and internal shear flows for the Fuselage Root station ($z = 2.515$ m).

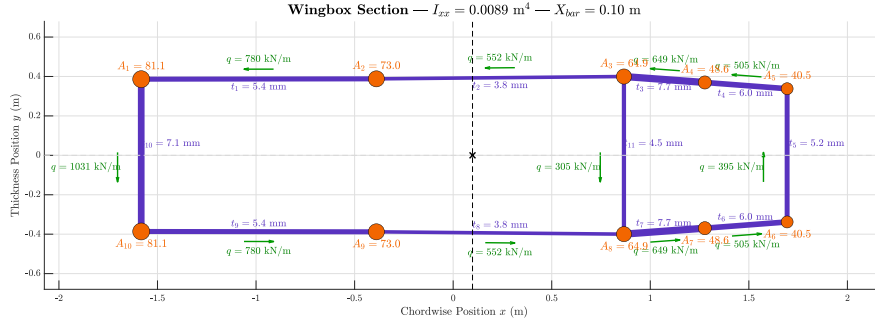


Figure 15: Fully stressed design cross-section and internal shear flows for the Aerodynamic Kink station ($z = 7.5$ m).

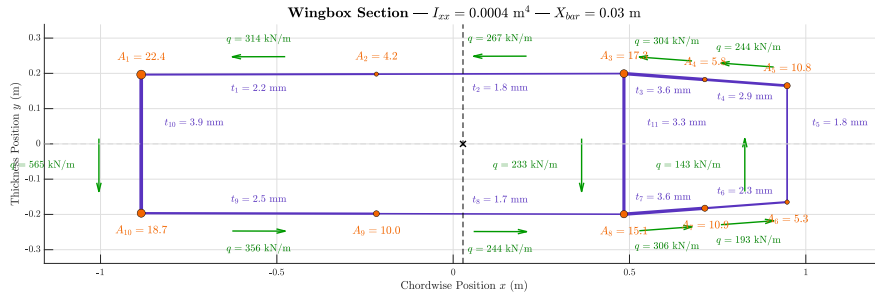


Figure 16: Fully stressed design cross-section and internal shear flows for the 75% Span station.

Note: In the given section plots, the tip is located behind, going into the page, and the root is located in front, coming out towards the reader. Therefore, the cut from the tip corresponds to a **negative shear force**, as observed in the structural graphs.

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	RF_σ	ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	85.93	-228.31	1.058	1	4.76	144.90	690.0	1.000
2	73.56	-234.90	1.028	2	3.72	144.90	539.7	1.000
3	68.65	-241.50	1.000	3	8.03	80.75	648.4	1.794
4	52.23	-222.35	1.086	4	6.81	80.43	547.4	1.801
5	45.59	-202.46	1.193	5	6.91	67.63	467.1	2.143
6	45.01	202.46	1.193	6	6.79	80.43	546.3	1.801
7	52.55	222.35	1.086	7	8.02	80.75	648.0	1.794
8	67.56	241.50	1.000	8	3.70	144.90	537.1	1.000
9	76.51	234.90	1.028	9	4.78	144.90	693.4	1.000
10	84.25	228.31	1.058	10	5.94	144.90	860.7	1.000
				11	4.21	59.99	252.8	2.415

Table 8: Optimal Sizing Results for Section 1: Fuselage Root ($m_w = 0.1243 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	RF_σ	ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	81.08	-233.40	1.035	1	5.38	144.90	779.7	1.000
2	72.97	-234.46	1.030	2	3.81	144.90	552.3	1.000
3	64.86	-241.50	1.000	3	7.66	84.81	649.3	1.709
4	48.65	-223.36	1.081	4	5.98	84.48	504.9	1.715
5	40.54	-204.23	1.182	5	5.23	75.44	394.8	1.921
6	40.54	204.23	1.182	6	5.98	84.48	504.9	1.715
7	48.65	223.36	1.081	7	7.66	84.81	649.3	1.709
8	64.86	241.50	1.000	8	3.81	144.90	552.3	1.000
9	72.97	234.46	1.030	9	5.38	144.90	779.7	1.000
10	81.08	233.40	1.035	10	7.11	144.90	1031.3	1.000
				11	4.53	67.44	305.2	2.149

Table 9: Optimal Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1080 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	RF_σ	ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	22.45	-238.03	1.015	1	2.17	144.90	314.2	1.000
2	4.21	-239.77	1.007	2	1.84	144.90	267.0	1.000
3	17.27	-241.50	1.000	3	3.57	85.17	304.2	1.701
4	5.77	-220.94	1.093	4	2.88	84.85	244.5	1.708
5	10.81	-199.92	1.208	5	1.83	78.31	143.2	1.850
6	5.35	199.92	1.208	6	2.28	84.85	193.3	1.708
7	10.85	220.94	1.093	7	3.59	85.17	305.6	1.701
8	15.12	241.50	1.000	8	1.68	144.90	244.0	1.000
9	9.97	239.77	1.007	9	2.45	144.90	355.9	1.000
10	18.71	238.02	1.015	10	3.89	144.90	564.5	1.000
				11	3.33	69.89	232.7	2.073

Table 10: Optimal Sizing Results for Section 3: 75% Span ($m_w = 0.0239 \text{ m}^2$)

Total Configuration Structural Weight Metric:

Evaluating the sum of the optimal cross-sectional areas from the three target stations ($0.1243 \text{ m}^2 + 0.1080 \text{ m}^2 + 0.0239 \text{ m}^2$), the global weight proxy for this specific structural arrangement results in $\mathbf{m_{w,total} = 0.2562 \text{ m}^2}$.

It is important to note that the algebraic signs of the shear stresses (τ) are not strictly indicative of failure modes, as they primarily depend on the assumed direction of the shear flow contour defined during the formulation of the multi-cell wingbox problem. Therefore, their magnitudes are enough to dictate the structural integrity against yielding. Conversely, the signs of the normal stresses (σ) in the booms are physically significant and highly consistent with the expected bending behavior of a cantilevered wing: the booms located on the upper skin exhibit negative values, indicating that the top of the wingbox is undergoing compression, while the lower booms would reflect tension, perfectly aligning with the upward aerodynamic lift loading.

Structurally, the superiority of the [15%, 53%, 65%] configuration lies in its ability to maximize the enclosed cross sectional area of the torsion box while leveraging the regions of maximum airfoil thickness. Advancing the front spar to 15% of the chord significantly increases both the bending inertia I_{xx} and the torsional stiffness without demanding excessively thick skin panels. Furthermore, it is critical to note that the selected airfoils inherently produce relatively flat wingboxes, characterized by a small vertical depth. This lack of structural height severely penalizes the section's moment of inertia, which directly dictates the topology of the problem: it forces the optimizer to concentrate the load-bearing material, such as placing the mid-spar at 53%, near the point of maximum thickness to efficiently withstand shear and bending. Simultaneously, the rear spar is constrained to 65% to avoid the drastic loss of geometric height towards the trailing edge, which would yield a highly inefficient structure and a severe weight penalty.

5 FEM Analysis

In this section, a finite element method (FEM) analysis using Abaqus/CAE will be presented to validate the analytical stress solutions obtained previously for the optimized wingbox configuration, which features spars located at 15%, 53%, and 60% of the chord. Specifically, a simplified simulation will be performed by modeling the wingbox segment between the fuselage intersection (Section 1) and the 75% wingspan station (Section 3). This model will assume a linear spanwise variation for both skin thicknesses and boom areas. Furthermore, the applied load distribution will be reduced to an equivalent force system applied at Section 3, designed to produce identical shear loads, bending moments, and torsional moments at the clamped root (Section 1).

5.1 Geometrical Modeling

First, the entire geometry must be modeled in the Part module, including the skin, spars, and the booms resulting from the axial load idealization. Thin-walled elements will be modeled as Shells (2D), using the Loft function to build the swept and tapered wingbox. To do this, two planar wire sections are built using the streamwise dimensions of stations 1 and 3, which are then lofted along the elastic axis, defined by its sweep angle of approximately 29.2° . Similarly, the booms are built directly onto this geometry using the point-to-point Wire (1D) function. It is important to note that the panels and the booms were modeled as separate parts to maintain precise property assignment, and were subsequently coupled in the *Assembly* module.

5.2 Assembly

In order to build the finite element model in Abaqus, the structural components were instantiated and positioned within the Assembly module. The assembly combines two distinct parts: the aerodynamic outer skin of the wing box, modeled as a 3D deformable shell, and the internal wireframe structure (spars and booms), defined as a 3D wire feature. Both components are illustrated in the figures below (see **Fig 18**). To ensure full kinematic coupling and a realistic transfer of loads between the internal structure and the outer skin, the two instances are subsequently connected using a tie constraint.

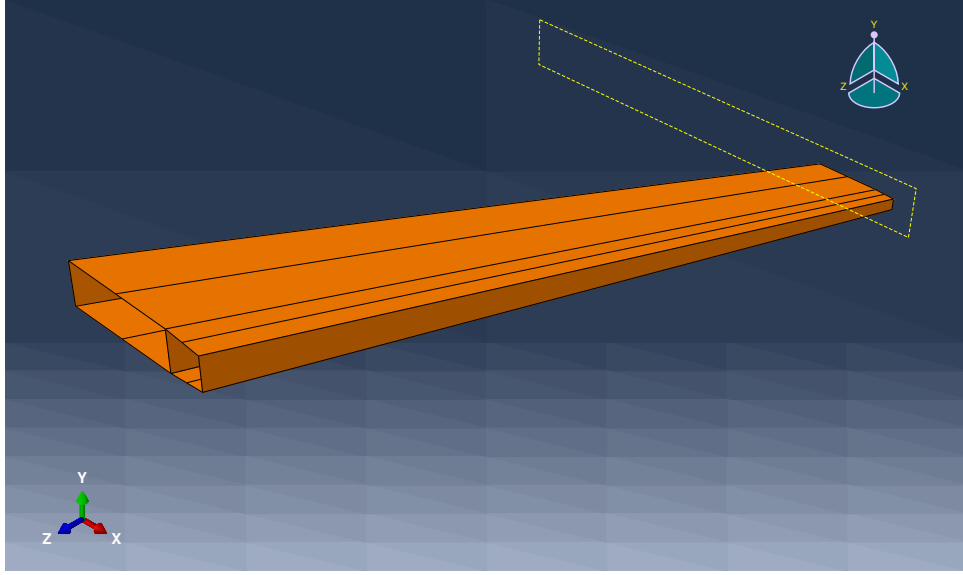


Figure 17: Isometric view of the optimum wingbox configuration modeled in Abaqus/CAE.

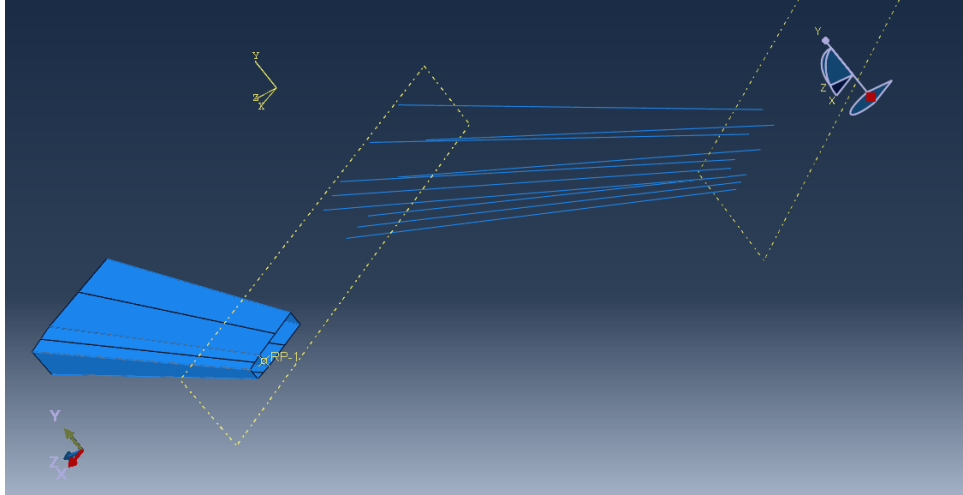


Figure 18: Implementation of two-part assembly consisting on wingbox deformable shell and wireframe.

5.3 Properties

Regarding the assignment of structural properties, the FE model has been simplified by defining two distinct element types to accurately capture the structural behavior of the wing-box. On one hand, **2D Shell Elements** were utilized for the outer skin and spar webs, modeled as homogeneous shell sections with their respective specified thicknesses. These elements are ideal for carrying the in-plane shear flows and distributed aerodynamic loads. On the other hand, **1D Beam Elements** were chosen for the booms, which represent the concentrated areas of stringers and spar caps. To implement this in the software, they were geometrically idealized with a circular profile, where the radius was mathematically calculated to yield a cross-sectional area exactly matching the analytical sizing results.

It is highly worth noting the structural implications of this choice: theoretically, **1D Truss Elements** would be the most accurate representation for the booms, as the analytical Fully Stressed Design (FSD) paradigm assumes they carry only pure axial loads. In contrast, 1D Beam Elements possess bending and shear stiffness, which introduces secondary stresses into the model. However, truss elements proved to be incompatible with the generated 3D mesh topology and the node connectivity required by the shell elements. Consequently, 1D Beams were adopted as the most robust and practical modeling compromise.

Finally, regarding the spanwise property distribution, the optimal thicknesses and areas obtained analytically do not exhibit a clear monotonic trend that would allow for a reliable continuous interpolation along the wingspan. To solve this, each structural component in the FE model was assigned the average value calculated from the three analyzed control stations (Root, Kink, and 75% Span). All elements were assigned the standard material properties for **2024-T3 Aluminium**, as previously introduced in the structure idealization section.

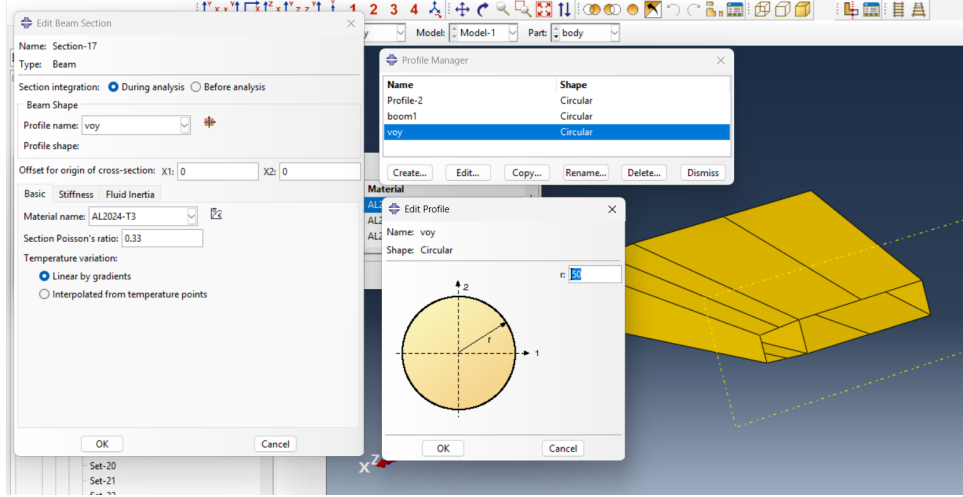


Figure 19: Beam section assignment through circular profile.

5.4 Interaction

Because the geometry was generated as separate parts, the implementation of constraints was required to dictate how both elements behave under the application of loads. Firstly, a surface-based Tie constraint was used to enforce kinematic continuity between the different element types, selecting the stiffer panels as the main surface, while the booms were designated as the secondary region.

Similarly, a kinematic coupling constraint was required at Section 3 to avoid stress singularities and localized distortions. To that effect, a Reference Point (RP) was created at the cross-section's elastic axis, and such constraint was used to link this point to the entire perimeter, ensuring that any shear or torsional loads are applied as a statically equivalent force system.

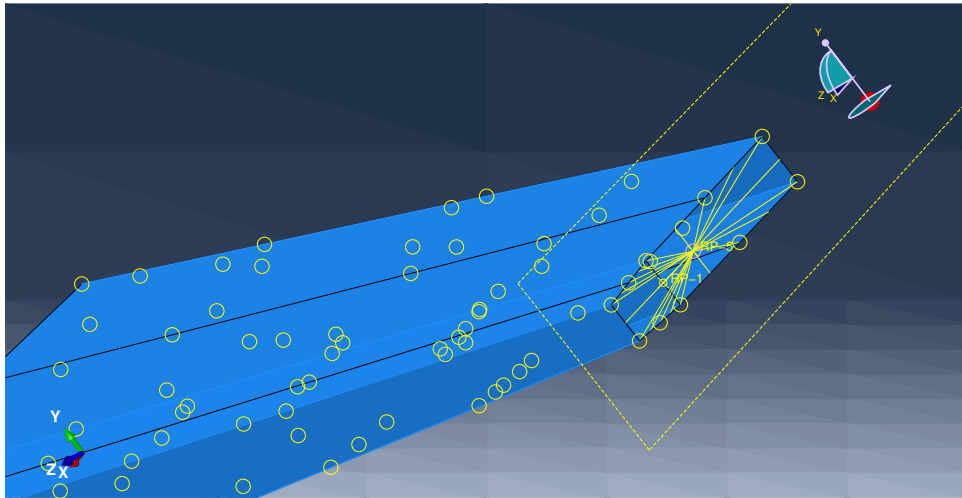


Figure 20: Implementation of Tie and Coupling constraints at the tip of the wingbox.

5.5 Load Implementation

As outlined above, Saint-Venant's principle will be applied to substitute the true distributed structural load with a statically equivalent system at the wing tip (Section 3). To mitigate the local artificial stress concentrations typically induced by this idealization at the point of application, a kinematic coupling constraint is implemented. This coupling guarantees a smooth distribution of forces across the interface nodes, thereby preventing unrealistic stress and strain peaks while ensuring a highly accurate stress field at the clamped root (Section 1). Based on the preceding analytical calculations, the equivalent concentrated loads required for the Abaqus/CAE simulation and expressed in the global are:

$$\begin{cases} V = V_1 = 748.89 \text{ kN} \\ M_x = M_{x_1} - V_1 \cdot \left(0.75 \cdot \frac{b}{2} - \frac{y_R}{2}\right) = -554.40 \text{ kN} \cdot \text{m} \\ M_z = M_{z_1} + V_1 \cdot \left(0.75 \cdot \frac{b}{2} - \frac{y_R}{2}\right) \cdot \tan(\Lambda_{el}) = 8612.11 \text{ kN} \cdot \text{m} \end{cases}$$

After appropriately defining a Static, General Step in the corresponding module, such loads can be applied directly as concentrated forces and moments to the Reference Point. For simplicity, non-linear geometric effects (Nlgeom) were disabled under the assumption of operation within the elastic regime, therefore matching the analytical formulation presented above. Additionally, the boundary conditions representing the wing attachment to the fuselage must be implemented at Section 1. In order to model a fully rigid cantilever joint, the ENCASTRE option was selected ($U_1 = U_2 = U_3 = UR_1 = UR_2 = UR_3 = 0$), constraining all six degrees of freedom for the nodes at the root and preventing any translation or rotation.

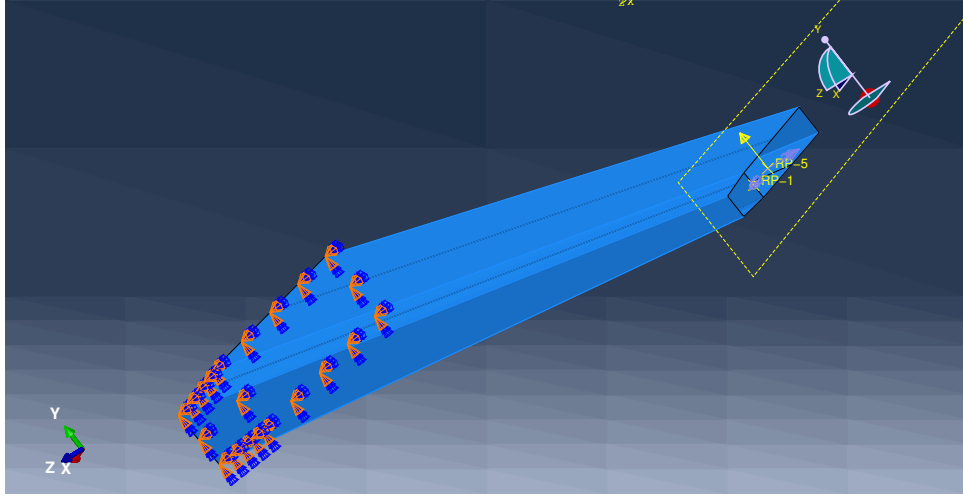
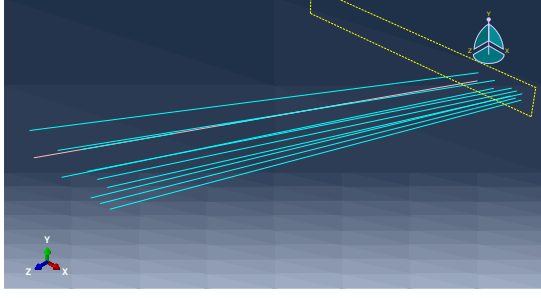


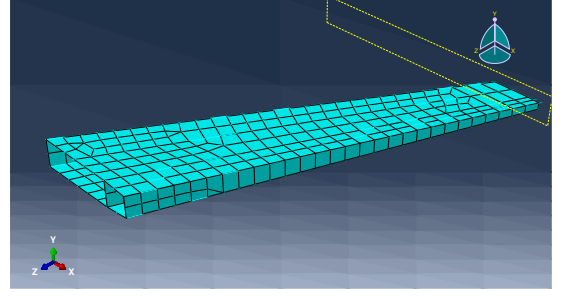
Figure 21: Boundary condition and loading scheme.

5.6 Mesh

Finally, prior to performing the simulation, both parts must be meshed independently, since they rely on different numerical formulations. Therefore, the skin and spar webs will be modeled as standard 2D elements (i.e., S4R, which is a 4-node shell), while the booms will be assigned B31 elements (Beams). However, ideally we should use T3D2 elements (Truss), which only carry axial loads, but they cannot be used with a standard formulation. Moreover, note that the quality of the mesh may be limited due to the 1000-node limit of the Abaqus Student Edition, which prevented convergence and mesh sensitivity analyses.



(a) Booms (B31).

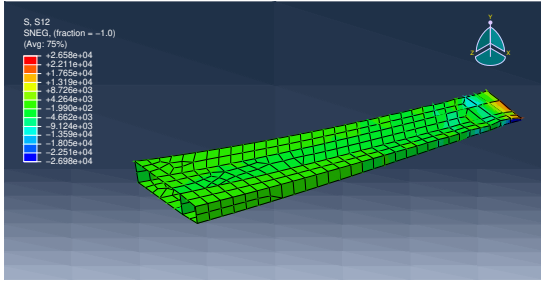


(b) Panels (S4R).

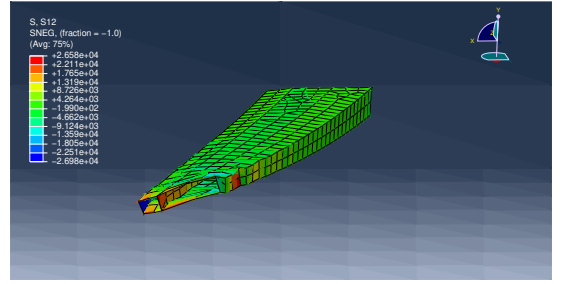
Figure 22: Independent meshing process of booms and panels.

5.7 Discussion of Results

The numerical analysis provided valuable insights to validate the analytical solution. For the panels, the finite element shear stress distribution at the clamped root aligns with the analytical predictions, yielding a magnitude of approximately 100 MPa. Additionally, the contour plots reveal that the aft chordwise panels near the tip experience higher shear stresses. This corroborates the analytical sizing, which required increased thickness in those specific regions. However, it must be noted that the peak stresses observed directly at the tip are likely exaggerated by localized artificial distortions resulting from the concentrated application of the equivalent loads. Similarly, the stress distribution in the booms



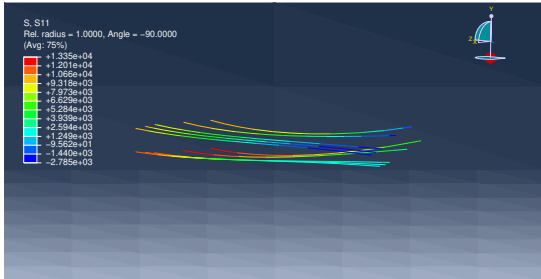
(a) Isometric view.



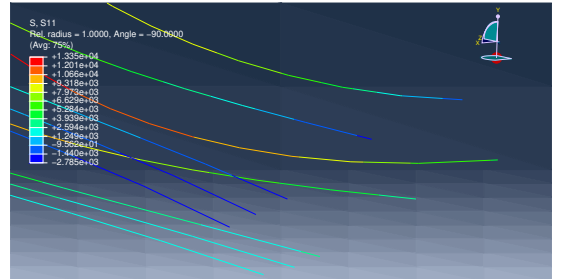
(b) Side view.

Figure 23: In-plane shear stress (S_{12}) contours on the shells.

is consistent with the analytical predictions to some extent, with compressive stresses in the upper booms and tensile stresses in the lower booms due to the bending moment induced by the applied load distribution. However, at the root section the entire cross-section is in tension, which is not physically consistent with the expected bending behavior. This suggests a limitation in the modeling approach or the implementation process.



(a) Isometric view.



(b) Zoomed view.

Figure 24: Axial stress (S_{11}) contours on the booms.

However, notable differences are observed in both the magnitude and distribution of the stress results, which can be attributed to several factors. These include limitations in mesh quality and resolution, as well as simplifications inherent to the analytical model, such as the neglect of inertial coupling effects under the assumption of the cross-section being symmetric.

6 Appendix

6.1 Project Deliverables

Report Deliverable	Task Description	Quick Link
Wing Stress Characteristic Plots	Load envelopes (V, M_x, M_z) and section stress distribution.	<i>Go to Section</i>
Wing Box Sizing Algorithm	Description of the iterative solver and optimization constraints.	<i>Go to Section</i>
Sizing Results (FSD)	Final A and t tables for Root, Kink, and 75% Span.	<i>Go to Section</i>
Configuration Weight	Weight proxy $\sum m_w$ and DoE results discussion.	<i>Go to Section</i>
FEM Description	Optional structural verification via Finite Element Method.	<i>Go to Section</i>

6.2 Non-optimal Configurations Sizing Results

CONFIGURATION: CASE 1 ($\eta_f = 0.15$, $\eta_m = 0.47$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	87.43	-216.09	1.118	1	4.73	144.90	685.9	1.000
2	71.74	-228.49	1.057	2	3.76	144.90	545.2	1.000
3	69.31	-241.50	1.000	3	7.38	91.18	673.0	1.589
4	50.79	-217.27	1.112	4	6.38	90.69	578.3	1.598
5	44.71	-191.62	1.260	5	5.95	84.87	504.8	1.707
6	43.73	191.62	1.260	6	6.36	90.69	576.7	1.598
7	51.53	217.27	1.112	7	7.38	91.18	672.8	1.589
8	67.00	241.50	1.000	8	3.73	144.90	540.2	1.000
9	77.57	228.49	1.057	9	4.78	144.90	692.3	1.000
10	83.97	216.09	1.118	10	5.85	144.90	848.1	1.000
				11	3.77	72.08	271.6	2.010

Table 11: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1241$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	80.57	-218.69	1.104	1	5.33	144.90	772.2	1.000
2	72.51	-234.22	1.031	2	3.78	144.90	547.7	1.000
3	64.45	-241.50	1.000	3	7.68	95.73	734.9	1.514
4	48.34	-226.89	1.064	4	6.19	95.24	589.9	1.521
5	40.28	-201.37	1.199	5	5.15	93.72	482.7	1.546
6	40.28	201.37	1.199	6	6.19	95.24	589.9	1.521
7	48.34	226.89	1.064	7	7.68	95.73	734.9	1.514
8	64.45	241.50	1.000	8	3.78	144.90	547.7	1.000
9	72.51	234.22	1.031	9	5.33	144.90	772.2	1.000
10	80.57	218.69	1.104	10	6.93	144.90	1005.1	1.000
				11	4.97	80.67	400.6	1.796

Table 12: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1096$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.39	-223.84	1.079	1	2.42	144.90	350.4	1.000
2	8.47	-232.50	1.039	2	1.78	144.90	258.3	1.000
3	17.04	-241.50	1.000	3	2.70	94.73	256.1	1.530
4	3.68	-215.20	1.122	4	2.32	94.25	219.0	1.537
5	8.45	-188.01	1.285	5	1.52	95.35	144.6	1.520
6	3.90	188.01	1.285	6	1.90	94.25	178.9	1.537
7	9.27	215.20	1.122	7	2.87	94.73	272.2	1.530
8	14.76	241.50	1.000	8	1.72	144.90	248.6	1.000
9	11.67	232.50	1.039	9	2.59	144.90	375.6	1.000
10	19.99	223.84	1.079	10	4.04	144.90	585.0	1.000
				11	2.32	81.97	190.4	1.768

Table 13: Sizing Results for Section 3: 75% Span ($m_w = 0.0233$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2570$ m².

CONFIGURATION: CASE 2 ($\eta_f = 0.15$, $\eta_m = 0.47$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	84.68	-216.25	1.117	1	4.51	144.90	654.2	1.000
2	72.23	-228.59	1.056	2	3.57	144.90	517.2	1.000
3	67.89	-241.50	1.000	3	6.88	94.32	649.0	1.536
4	51.05	-208.07	1.161	4	5.99	93.70	560.9	1.546
5	46.63	-172.18	1.403	5	5.20	94.97	494.2	1.526
6	43.72	172.18	1.403	6	5.94	93.70	556.7	1.546
7	51.90	208.07	1.161	7	6.85	94.32	646.3	1.536
8	68.37	241.50	1.000	8	3.56	144.90	515.4	1.000
9	76.96	228.59	1.056	9	4.56	144.90	661.4	1.000
10	80.64	216.25	1.117	10	5.56	144.90	806.2	1.000
				11	3.42	78.25	268.0	1.852

Table 14: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1246$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	78.17	-227.85	1.060	1	5.01	144.90	726.9	1.000
2	70.35	-241.50	1.000	2	3.52	144.90	510.9	1.000
3	62.53	-241.50	1.000	3	7.45	98.63	734.9	1.469
4	46.90	-226.62	1.066	4	6.13	98.02	601.1	1.478
5	39.08	-189.02	1.278	5	4.91	103.47	508.1	1.400
6	39.08	189.02	1.278	6	6.13	98.02	601.1	1.478
7	46.90	226.62	1.066	7	7.45	98.63	734.9	1.469
8	62.53	241.50	1.000	8	3.52	144.90	510.9	1.000
9	70.35	241.50	1.000	9	5.01	144.90	726.9	1.000
10	78.17	227.85	1.060	10	6.56	144.90	951.2	1.000
				11	4.94	86.86	429.5	1.668

Table 15: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1101$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.45	-223.99	1.078	1	2.64	144.90	382.6	1.000
2	10.66	-232.59	1.038	2	1.83	144.90	265.2	1.000
3	17.45	-241.50	1.000	3	2.46	96.98	238.7	1.494
4	1.01	-205.55	1.175	4	2.38	96.37	228.9	1.504
5	8.42	-168.07	1.437	5	1.55	104.48	161.9	1.387
6	2.79	168.07	1.437	6	1.91	96.37	184.1	1.504
7	7.32	205.55	1.175	7	2.63	96.98	255.3	1.494
8	15.28	241.50	1.000	8	1.77	144.90	257.0	1.000
9	14.28	232.59	1.038	9	2.86	144.90	414.3	1.000
10	19.47	223.99	1.078	10	4.28	144.90	620.8	1.000
				11	1.98	87.50	173.2	1.656

Table 16: Sizing Results for Section 3: 75% Span ($m_w = 0.0242$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 2 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2589$ m².

CONFIGURATION: CASE 3 ($\eta_f = 0.15$, $\eta_m = 0.47$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	84.69	-216.44	1.116	1	4.25	144.90	616.1	1.000
2	73.30	-228.71	1.056	2	3.36	144.90	487.1	1.000
3	67.95	-241.50	1.000	3	6.86	95.29	653.9	1.521
4	50.90	-197.92	1.220	4	6.10	94.54	576.4	1.533
5	44.84	-150.49	1.605	5	5.12	102.43	524.5	1.415
6	43.70	150.49	1.605	6	6.08	94.54	575.1	1.533
7	51.53	197.92	1.220	7	6.86	95.29	653.5	1.521
8	67.57	241.50	1.000	8	3.35	144.90	486.1	1.000
9	75.45	228.71	1.056	9	4.27	144.90	618.8	1.000
10	83.05	216.44	1.116	10	5.22	144.90	757.1	1.000
				11	3.58	81.83	293.0	1.771

Table 17: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1280$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	76.03	-237.82	1.015	1	4.95	144.90	716.9	1.000
2	68.42	-241.50	1.000	2	3.45	144.90	499.4	1.000
3	60.82	-241.50	1.000	3	7.33	99.26	727.6	1.460
4	45.62	-225.21	1.072	4	6.08	98.52	599.3	1.471
5	38.01	-172.88	1.397	5	4.69	110.26	517.2	1.314
6	38.01	172.88	1.397	6	6.08	98.52	599.3	1.471
7	45.62	225.21	1.072	7	7.33	99.26	727.6	1.460
8	60.82	241.50	1.000	8	3.45	144.90	499.4	1.000
9	68.42	241.50	1.000	9	4.95	144.90	716.9	1.000
10	76.03	237.82	1.015	10	6.50	144.90	942.6	1.000
				11	4.82	90.20	434.8	1.606

Table 18: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1118$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.31	-224.16	1.077	1	2.68	144.90	388.8	1.000
2	11.26	-232.71	1.038	2	1.82	144.90	263.2	1.000
3	17.24	-241.50	1.000	3	2.66	97.01	257.6	1.494
4	1.04	-195.03	1.238	4	2.57	96.26	247.8	1.505
5	8.70	-146.16	1.652	5	1.69	110.72	186.9	1.309
6	1.95	146.17	1.652	6	2.08	96.26	200.6	1.505
7	8.12	195.03	1.238	7	2.85	97.01	276.5	1.494
8	15.78	241.50	1.000	8	1.83	144.90	265.2	1.000
9	13.47	232.70	1.038	9	2.87	144.90	415.4	1.000
10	19.84	224.16	1.077	10	4.34	144.90	628.6	1.000
				11	2.15	90.21	194.1	1.606

Table 19: Sizing Results for Section 3: 75% Span ($m_w = 0.0257$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 3 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2655}$ m².

CONFIGURATION: CASE 4 ($\eta_f = 0.15$, $\eta_m = 0.50$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	85.75	-221.56	1.090	1	4.69	144.90	680.1	1.000
2	72.27	-231.35	1.044	2	3.69	144.90	535.7	1.000
3	70.77	-241.50	1.000	3	7.84	86.42	677.9	1.677
4	51.11	-219.52	1.100	4	6.76	86.01	581.0	1.685
5	44.90	-196.47	1.229	5	6.57	76.88	504.8	1.885
6	44.54	196.47	1.229	6	6.75	86.01	580.4	1.685
7	52.14	219.52	1.100	7	7.86	86.42	679.3	1.677
8	67.44	241.50	1.000	8	3.66	144.90	530.0	1.000
9	77.85	231.35	1.044	9	4.73	144.90	685.6	1.000
10	82.85	221.56	1.090	10	5.82	144.90	844.2	1.000
				11	4.35	66.67	290.0	2.173

Table 20: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1250$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	80.81	-223.41	1.081	1	5.35	144.90	776.1	1.000
2	72.73	-236.55	1.021	2	3.78	144.90	548.1	1.000
3	64.65	-241.50	1.000	3	7.69	91.17	700.6	1.589
4	48.49	-228.65	1.056	4	6.10	90.76	553.7	1.597
5	40.41	-205.71	1.174	5	5.17	85.76	443.5	1.690
6	40.41	205.71	1.174	6	6.10	90.76	553.7	1.597
7	48.49	228.65	1.056	7	7.69	91.17	700.6	1.589
8	64.65	241.50	1.000	8	3.78	144.90	548.1	1.000
9	72.73	236.55	1.021	9	5.35	144.90	776.1	1.000
10	80.81	223.41	1.081	10	7.01	144.90	1015.4	1.000
				11	4.88	75.13	366.9	1.929

Table 21: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1087$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.50	-230.25	1.049	1	2.28	144.90	330.7	1.000
2	6.93	-235.77	1.024	2	1.75	144.90	254.2	1.000
3	17.04	-241.50	1.000	3	3.00	90.42	271.2	1.603
4	3.97	-217.78	1.109	4	2.56	90.01	230.7	1.610
5	9.83	-193.39	1.249	5	1.62	87.63	141.8	1.654
6	4.31	193.39	1.249	6	2.01	90.01	180.7	1.610
7	9.51	217.78	1.109	7	3.07	90.42	277.7	1.603
8	15.14	241.50	1.000	8	1.65	144.90	239.4	1.000
9	11.56	235.77	1.024	9	2.53	144.90	367.0	1.000
10	19.13	230.25	1.049	10	3.95	144.90	573.2	1.000
				11	2.73	76.68	209.6	1.890

Table 22: Sizing Results for Section 3: 75% Span ($m_w = 0.0234$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 4 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2571}$ m².

CONFIGURATION: CASE 5 ($\eta_f = 0.15$, $\eta_m = 0.50$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	86.92	-221.74	1.089	1	4.52	144.90	654.7	1.000
2	71.83	-231.46	1.043	2	3.56	144.90	515.9	1.000
3	68.55	-241.50	1.000	3	7.15	90.79	649.1	1.596
4	50.93	-210.02	1.150	4	6.20	90.25	559.8	1.606
5	44.93	-176.55	1.368	5	5.55	88.86	493.5	1.631
6	43.81	176.55	1.368	6	6.18	90.25	558.1	1.606
7	51.51	210.02	1.150	7	7.14	90.79	648.5	1.596
8	66.48	241.50	1.000	8	3.52	144.90	511.0	1.000
9	77.68	231.46	1.043	9	4.56	144.90	661.2	1.000
10	83.42	221.74	1.089	10	5.63	144.90	815.8	1.000
				11	3.65	74.52	271.6	1.945

Table 23: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1250$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	78.40	-232.92	1.037	1	5.03	144.90	728.7	1.000
2	70.56	-241.50	1.000	2	3.51	144.90	509.1	1.000
3	62.72	-241.50	1.000	3	7.52	95.33	717.0	1.520
4	47.04	-228.29	1.058	4	6.13	94.79	581.4	1.529
5	39.20	-193.23	1.250	5	4.97	97.67	485.8	1.484
6	39.20	193.23	1.250	6	6.13	94.79	581.4	1.529
7	47.04	228.29	1.058	7	7.52	95.33	717.0	1.520
8	62.72	241.50	1.000	8	3.51	144.90	509.1	1.000
9	70.56	241.50	1.000	9	5.03	144.90	728.7	1.000
10	78.40	232.92	1.037	10	6.62	144.90	959.3	1.000
				11	4.98	83.19	414.3	1.742

Table 24: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1094$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.48	-230.41	1.048	1	2.36	144.90	341.8	1.000
2	8.16	-235.87	1.024	2	1.73	144.90	250.6	1.000
3	17.33	-241.50	1.000	3	2.80	93.89	263.3	1.543
4	4.65	-207.82	1.162	4	2.33	93.35	217.5	1.552
5	7.91	-172.89	1.397	5	1.54	98.91	152.7	1.465
6	3.79	172.89	1.397	6	1.97	93.35	183.8	1.552
7	10.02	207.82	1.162	7	3.01	93.89	282.4	1.543
8	15.64	241.50	1.000	8	1.73	144.90	250.4	1.000
9	9.05	235.87	1.024	9	2.43	144.90	351.7	1.000
10	21.58	230.41	1.048	10	4.05	144.90	587.3	1.000
				11	2.51	84.08	211.1	1.723

Table 25: Sizing Results for Section 3: 75% Span ($m_w = 0.0242$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 5 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2585$ m².

CONFIGURATION: CASE 6 ($\eta_f = 0.15$, $\eta_m = 0.50$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	RF_σ	ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	85.83	-221.95	1.088	1	4.28	144.90	621.1	1.000
2	73.02	-231.58	1.043	2	3.38	144.90	490.1	1.000
3	68.32	-241.50	1.000	3	7.01	92.51	648.7	1.566
4	50.79	-199.55	1.210	4	6.21	91.83	570.2	1.578
5	44.98	-154.31	1.565	5	5.29	97.65	516.4	1.484
6	43.63	154.31	1.565	6	6.19	91.83	568.6	1.578
7	51.42	199.55	1.210	7	7.01	92.51	648.0	1.566
8	67.76	241.50	1.000	8	3.37	144.90	488.5	1.000
9	75.93	231.58	1.043	9	4.31	144.90	624.7	1.000
10	83.76	221.95	1.088	10	5.30	144.90	768.7	1.000
				11	3.62	79.22	286.4	1.829

Table 26: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1278 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	RF_σ	ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	76.24	-241.50	1.000	1	4.92	144.90	713.6	1.000
2	68.61	-241.50	1.000	2	3.40	144.90	492.5	1.000
3	60.99	-241.50	1.000	3	7.42	96.73	717.4	1.498
4	45.74	-226.76	1.065	4	6.12	96.06	587.5	1.508
5	38.12	-176.86	1.366	5	4.75	105.92	503.2	1.368
6	38.12	176.86	1.366	6	6.12	96.06	587.5	1.508
7	45.74	226.76	1.065	7	7.42	96.73	717.4	1.498
8	60.99	241.50	1.000	8	3.40	144.90	492.5	1.000
9	68.61	241.50	1.000	9	4.92	144.90	713.6	1.000
10	76.24	241.50	1.000	10	6.53	144.90	945.8	1.000
				11	4.93	87.75	432.7	1.651

Table 27: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1111 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	RF_σ	ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	22.50	-230.59	1.047	1	2.53	144.90	367.3	1.000
2	10.41	-235.99	1.023	2	1.72	144.90	249.4	1.000
3	17.01	-241.50	1.000	3	2.69	94.64	254.8	1.531
4	1.96	-196.96	1.226	4	2.51	93.97	236.2	1.542
5	9.28	-150.36	1.606	5	1.59	106.56	169.3	1.360
6	3.34	150.36	1.606	6	2.06	93.97	193.4	1.542
7	8.85	196.96	1.226	7	2.93	94.64	277.0	1.531
8	15.33	241.50	1.000	8	1.74	144.90	252.3	1.000
9	11.42	235.99	1.023	9	2.63	144.90	381.5	1.000
10	21.21	230.60	1.047	10	4.25	144.90	616.2	1.000
				11	2.30	87.97	202.4	1.647

Table 28: Sizing Results for Section 3: 75% Span ($m_w = 0.0254 \text{ m}^2$)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 6 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2643 \text{ m}^2}$.

★ OPTIMAL CONFIGURATION: CASE 7 ($\eta_f = 0.15$,
 $\eta_m = 0.53$, $\eta_r = 0.65$) ★

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	85.93	-228.31	1.058	1	4.76	144.90	690.0	1.000
2	73.56	-234.90	1.028	2	3.72	144.90	539.7	1.000
3	68.65	-241.50	1.000	3	8.03	80.75	648.4	1.794
4	52.23	-222.35	1.086	4	6.81	80.43	547.4	1.801
5	45.59	-202.46	1.193	5	6.91	67.63	467.1	2.143
6	45.01	202.46	1.193	6	6.79	80.43	546.3	1.801
7	52.55	222.35	1.086	7	8.02	80.75	648.0	1.794
8	67.56	241.50	1.000	8	3.70	144.90	537.1	1.000
9	76.51	234.90	1.028	9	4.78	144.90	693.4	1.000
10	84.25	228.31	1.058	10	5.94	144.90	860.7	1.000
				11	4.21	59.99	252.8	2.415

Table 29: Optimal Sizing Results for Section 1: Fuselage Root ($m_w = 0.1243$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	81.08	-229.13	1.054	1	5.37	144.90	778.2	1.000
2	72.97	-241.50	1.000	2	3.77	144.90	546.0	1.000
3	64.86	-241.50	1.000	3	7.74	85.65	662.7	1.692
4	48.65	-230.75	1.047	4	6.02	85.32	513.5	1.698
5	40.54	-210.98	1.145	5	5.24	76.25	399.8	1.900
6	40.54	210.98	1.145	6	6.02	85.32	513.5	1.698
7	48.65	230.75	1.047	7	7.74	85.65	662.7	1.692
8	64.86	241.50	1.000	8	3.77	144.90	546.0	1.000
9	72.97	241.50	1.000	9	5.37	144.90	778.2	1.000
10	81.08	229.13	1.054	10	7.07	144.90	1025.1	1.000
				11	4.87	68.13	331.8	2.127

Table 30: Optimal Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1079$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.45	-238.03	1.015	1	2.17	144.90	314.2	1.000
2	4.21	-241.50	1.000	2	1.84	144.90	267.0	1.000
3	17.27	-241.50	1.000	3	3.57	85.17	304.2	1.701
4	5.77	-220.94	1.093	4	2.88	84.85	244.5	1.708
5	10.81	-199.92	1.208	5	1.83	78.31	143.2	1.850
6	5.35	199.92	1.208	6	2.28	84.85	193.3	1.708
7	10.85	220.94	1.093	7	3.59	85.17	305.6	1.701
8	15.12	241.50	1.000	8	1.68	144.90	244.0	1.000
9	9.97	241.50	1.000	9	2.45	144.90	355.9	1.000
10	18.71	238.02	1.015	10	3.89	144.90	564.5	1.000
				11	3.33	69.89	232.7	2.073

Table 31: Optimal Sizing Results for Section 3: 75% Span ($m_w = 0.0239$ m²)

Optimal Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations yields the global minimum weight proxy of $\mathbf{m}_{\mathbf{w},\text{total}} = \mathbf{0.2562\ m^2}$.

CONFIGURATION: CASE 8 ($\eta_f = 0.15$, $\eta_m = 0.53$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	86.53	-228.51	1.057	1	4.55	144.90	659.0	1.000
2	72.66	-235.02	1.028	2	3.55	144.90	515.3	1.000
3	68.66	-241.50	1.000	3	7.35	86.82	638.5	1.669
4	51.65	-212.50	1.136	4	6.32	86.35	546.1	1.678
5	44.60	-181.94	1.327	5	5.83	81.91	477.8	1.769
6	44.45	181.94	1.327	6	6.32	86.35	545.9	1.678
7	51.56	212.50	1.136	7	7.35	86.82	638.1	1.669
8	67.62	241.50	1.000	8	3.54	144.90	512.8	1.000
9	75.43	235.02	1.028	9	4.57	144.90	662.0	1.000
10	84.99	228.51	1.057	10	5.69	144.90	825.4	1.000
				11	3.75	70.00	262.7	2.070

Table 32: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1248$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	78.66	-239.08	1.010	1	5.04	144.90	730.5	1.000
2	70.79	-241.50	1.000	2	3.49	144.90	506.6	1.000
3	62.93	-241.50	1.000	3	7.64	91.55	699.2	1.583
4	47.19	-230.30	1.049	4	6.17	91.09	561.6	1.591
5	39.33	-198.34	1.218	5	5.09	90.87	462.8	1.595
6	39.33	198.34	1.218	6	6.17	91.09	561.6	1.591
7	47.19	230.30	1.049	7	7.64	91.55	699.2	1.583
8	62.93	241.50	1.000	8	3.49	144.90	506.6	1.000
9	70.79	241.50	1.000	9	5.04	144.90	730.5	1.000
10	78.66	239.08	1.010	10	6.68	144.90	968.7	1.000
				11	5.09	78.63	400.0	1.843

Table 33: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1088$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.44	-238.20	1.014	1	2.21	144.90	320.2	1.000
2	6.82	-241.50	1.000	2	1.67	144.90	242.6	1.000
3	17.13	-241.50	1.000	3	3.19	90.33	288.5	1.604
4	5.21	-210.60	1.147	4	2.63	89.87	236.4	1.612
5	9.29	-178.73	1.351	5	1.71	92.34	157.7	1.569
6	4.08	178.74	1.351	6	2.14	89.87	192.3	1.612
7	9.97	210.61	1.147	7	3.23	90.33	291.9	1.604
8	15.83	241.50	1.000	8	1.59	144.90	231.0	1.000
9	10.70	241.50	1.000	9	2.43	144.90	352.8	1.000
10	19.55	238.20	1.014	10	3.96	144.90	573.6	1.000
				11	3.04	79.77	242.1	1.816

Table 34: Sizing Results for Section 3: 75% Span ($m_w = 0.0244$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 8 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2579}$ m².

CONFIGURATION: CASE 9 ($\eta_f = 0.15$, $\eta_m = 0.53$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	85.79	-228.73	1.056	1	4.23	144.90	613.4	1.000
2	71.84	-235.16	1.027	2	3.32	144.90	481.5	1.000
3	71.92	-241.50	1.000	3	7.49	89.50	670.5	1.619
4	50.05	-201.63	1.198	4	6.66	88.91	591.7	1.630
5	43.12	-159.03	1.519	5	5.83	92.30	538.2	1.570
6	41.61	159.03	1.519	6	6.63	88.91	589.9	1.630
7	52.03	201.63	1.198	7	7.51	89.50	671.8	1.619
8	69.16	241.50	1.000	8	3.29	144.90	477.5	1.000
9	76.52	235.16	1.027	9	4.26	144.90	618.1	1.000
10	83.20	228.73	1.056	10	5.29	144.90	766.7	1.000
				11	4.27	76.11	324.7	1.904

Table 35: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1289$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	76.48	-241.50	1.000	1	4.90	144.90	710.3	1.000
2	68.83	-241.50	1.000	2	3.34	144.90	484.8	1.000
3	61.19	-241.50	1.000	3	7.58	93.93	711.6	1.543
4	45.89	-228.64	1.056	4	6.21	93.34	579.8	1.552
5	38.24	-181.69	1.329	5	4.88	100.92	492.6	1.436
6	38.24	181.69	1.329	6	6.21	93.34	579.8	1.552
7	45.89	228.64	1.056	7	7.58	93.93	711.6	1.543
8	61.19	241.50	1.000	8	3.34	144.90	484.8	1.000
9	68.83	241.50	1.000	9	4.90	144.90	710.3	1.000
10	76.48	241.50	1.000	10	6.56	144.90	950.4	1.000
				11	5.14	84.71	435.6	1.711

Table 36: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1105$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.20	-238.41	1.013	1	2.41	144.90	348.8	1.000
2	9.96	-241.50	1.000	2	1.61	144.90	234.1	1.000
3	16.60	-241.50	1.000	3	2.72	92.01	250.5	1.575
4	3.30	-199.33	1.212	4	2.39	91.42	218.9	1.585
5	9.27	-155.45	1.554	5	1.47	101.76	149.8	1.424
6	3.22	155.45	1.554	6	1.90	91.42	173.8	1.585
7	10.21	199.33	1.212	7	2.95	92.01	271.4	1.575
8	15.00	241.50	1.000	8	1.63	144.90	236.5	1.000
9	10.94	241.50	1.000	9	2.50	144.90	362.5	1.000
10	21.00	238.40	1.013	10	4.16	144.90	602.8	1.000
				11	2.45	85.17	208.8	1.701

Table 37: Sizing Results for Section 3: 75% Span ($m_w = 0.0249$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 9 yields a global weight proxy of $\mathbf{m}_{\mathbf{w},\text{total}} = \mathbf{0.2643}$ m².

Boom Sizing			
ID	A (cm ²)	σ (MPa)	RF_σ
1	82.57	-235.27	1.026
2	70.99	-238.46	1.013
3	66.76	-241.50	1.000
4	50.03	-217.22	1.112
5	44.18	-191.54	1.261
6	43.48	191.54	1.261
7	51.02	217.22	1.112
8	65.05	241.50	1.000
9	73.97	238.46	1.013
10	80.95	235.27	1.026

Panel Sizing				
ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	5.75	144.90	833.6	1.000
2	4.69	144.90	679.9	1.000
3	8.25	91.21	752.3	1.589
4	7.21	90.72	653.7	1.597
5	6.79	84.97	576.9	1.705
6	7.19	90.72	652.5	1.597
7	8.26	91.21	753.1	1.589
8	4.67	144.90	676.9	1.000
9	5.77	144.90	837.1	1.000
10	6.97	144.90	1010.0	1.000
11	3.07	71.34	218.8	2.031

Table 38: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1274 \text{ m}^2$)

Boom Sizing			
ID	A (cm ²)	σ (MPa)	RF_σ
1	81.43	-230.43	1.048
2	70.01	-236.05	1.023
3	65.34	-241.50	1.000
4	46.71	-218.64	1.105
5	39.90	-194.01	1.245
6	39.25	194.01	1.245
7	47.05	218.64	1.105
8	64.98	241.50	1.000
9	71.76	236.05	1.023
10	80.25	230.43	1.048

Panel Sizing				
ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	6.14	144.90	889.6	1.000
2	4.64	144.90	672.8	1.000
3	9.46	94.87	897.5	1.527
4	8.09	94.39	763.5	1.535
5	7.12	92.97	661.9	1.559
6	8.07	94.39	761.8	1.535
7	9.45	94.87	896.8	1.527
8	4.63	144.90	671.0	1.000
9	6.16	144.90	893.2	1.000
10	7.84	144.90	1135.9	1.000
11	5.46	79.07	431.8	1.832

Table 39: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1164 \text{ m}^2$)

Boom Sizing			
ID	A (cm ²)	σ (MPa)	RF_σ
1	22.27	-241.50	1.000
2	6.85	-241.50	1.000
3	16.19	-241.50	1.000
4	2.20	-214.47	1.126
5	7.56	-187.34	1.289
6	2.69	187.34	1.289
7	7.39	214.47	1.126
8	13.72	241.50	1.000
9	12.38	241.50	1.000
10	18.39	241.50	1.000

Panel Sizing				
ID	t (mm)	τ (MPa)	q (kN/m)	RF_τ
1	2.97	144.90	430.3	1.000
2	2.42	144.90	351.5	1.000
3	3.37	93.94	317.0	1.543
4	3.15	93.46	294.5	1.550
5	2.40	94.57	226.9	1.532
6	2.68	93.46	250.9	1.550
7	3.48	93.94	326.5	1.543
8	2.29	144.90	332.6	1.000
9	3.28	144.90	475.0	1.000
10	4.74	144.90	686.8	1.000
11	1.88	80.33	151.4	1.804

Table 40: Sizing Results for Section 3: 75% Span ($m_w = 0.0242 \text{ m}^2$)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 10 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2680 \text{ m}^2}$.

CONFIGURATION: CASE 11 ($\eta_f = 0.20$, $\eta_m = 0.47$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	82.55	-235.43	1.026	1	5.19	144.90	753.0	1.000
2	70.31	-238.56	1.012	2	4.21	144.90	610.5	1.000
3	66.51	-241.50	1.000	3	8.11	94.24	764.5	1.538
4	49.37	-208.01	1.161	4	7.23	93.63	677.3	1.548
5	43.87	-172.08	1.403	5	6.46	94.96	613.1	1.526
6	42.81	172.08	1.403	6	7.22	93.63	675.7	1.548
7	50.30	208.01	1.161	7	8.11	94.24	764.6	1.538
8	65.30	241.50	1.000	8	4.19	144.90	608.1	1.000
9	73.41	238.56	1.012	9	5.22	144.90	756.9	1.000
10	80.60	235.43	1.026	10	6.33	144.90	918.1	1.000
				11	3.76	77.26	290.4	1.875

Table 41: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1301$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	79.33	-234.30	1.031	1	6.16	144.90	892.3	1.000
2	71.40	-241.50	1.000	2	4.62	144.90	669.6	1.000
3	63.46	-241.50	1.000	3	8.29	97.63	809.2	1.484
4	47.60	-213.12	1.133	4	6.98	97.01	677.3	1.494
5	39.66	-177.72	1.359	5	5.71	102.59	585.7	1.412
6	39.66	177.72	1.359	6	6.98	97.01	677.3	1.494
7	47.60	213.12	1.133	7	8.29	97.63	809.2	1.484
8	63.46	241.50	1.000	8	4.62	144.90	669.6	1.000
9	71.40	241.50	1.000	9	6.16	144.90	892.3	1.000
10	79.33	234.30	1.031	10	7.82	144.90	1133.9	1.000
				11	4.03	84.97	342.0	1.705

Table 42: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1158$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.27	-241.50	1.000	1	3.05	144.90	442.6	1.000
2	7.47	-241.50	1.000	2	2.45	144.90	355.7	1.000
3	16.13	-241.50	1.000	3	3.24	96.08	311.0	1.508
4	2.23	-204.75	1.179	4	3.03	95.47	288.9	1.518
5	7.28	-167.38	1.443	5	2.22	103.62	230.1	1.398
6	2.41	167.38	1.443	6	2.61	95.47	249.6	1.518
7	7.51	204.75	1.179	7	3.37	96.08	323.8	1.508
8	13.75	241.50	1.000	8	2.35	144.90	341.0	1.000
9	12.23	241.50	1.000	9	3.33	144.90	483.3	1.000
10	18.78	241.50	1.000	10	4.84	144.90	702.2	1.000
				11	1.66	85.61	142.5	1.693

Table 43: Sizing Results for Section 3: 75% Span ($m_w = 0.0252$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 11 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2710$ m².

CONFIGURATION: CASE 12 ($\eta_f = 0.20$, $\eta_m = 0.47$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	82.47	-235.62	1.025	1	5.10	144.90	739.1	1.000
2	71.02	-238.67	1.012	2	4.13	144.90	599.2	1.000
3	67.74	-241.50	1.000	3	7.73	95.13	735.4	1.523
4	47.91	-197.85	1.221	4	6.96	94.39	657.2	1.535
5	40.79	-150.38	1.606	5	5.93	102.34	606.5	1.416
6	39.79	150.38	1.606	6	6.95	94.39	655.9	1.535
7	49.18	197.85	1.221	7	7.74	95.13	736.2	1.523
8	65.74	241.50	1.000	8	4.11	144.90	596.1	1.000
9	74.54	238.67	1.012	9	5.12	144.90	742.8	1.000
10	80.52	235.62	1.025	10	6.20	144.90	899.4	1.000
				11	3.36	80.66	271.1	1.796

Table 44: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1322$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	77.23	-241.50	1.000	1	5.83	144.90	844.8	1.000
2	69.51	-241.50	1.000	2	4.28	144.90	620.9	1.000
3	61.79	-241.50	1.000	3	8.49	98.15	833.2	1.476
4	46.34	-211.27	1.143	4	7.26	97.42	706.9	1.487
5	38.62	-162.13	1.490	5	5.73	109.33	626.1	1.325
6	38.62	162.13	1.490	6	7.26	97.42	706.9	1.487
7	46.34	211.27	1.143	7	8.49	98.15	833.2	1.476
8	61.79	241.50	1.000	8	4.28	144.90	620.9	1.000
9	69.51	241.50	1.000	9	5.83	144.90	844.8	1.000
10	77.23	241.50	1.000	10	7.51	144.90	1087.9	1.000
				11	4.72	88.11	415.9	1.645

Table 45: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1191$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.21	-241.50	1.000	1	2.86	144.90	415.1	1.000
2	8.23	-241.50	1.000	2	2.24	144.90	325.3	1.000
3	16.20	-241.50	1.000	3	3.42	96.05	328.2	1.509
4	1.14	-194.17	1.244	4	3.34	95.32	318.1	1.520
5	7.94	-145.49	1.660	5	2.42	109.86	265.9	1.319
6	1.51	145.49	1.660	6	2.89	95.32	275.8	1.520
7	7.06	194.17	1.244	7	3.52	96.05	337.9	1.509
8	14.83	241.50	1.000	8	2.21	144.90	320.1	1.000
9	12.69	241.50	1.000	9	3.16	144.90	458.7	1.000
10	18.24	241.50	1.000	10	4.54	144.90	658.2	1.000
				11	2.03	88.17	179.4	1.643

Table 46: Sizing Results for Section 3: 75% Span ($m_w = 0.0265$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 12 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2777}$ m².

CONFIGURATION: CASE 13 ($\eta_f = 0.20$, $\eta_m = 0.50$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	82.97	-241.50	1.000	1	5.74	144.90	831.6	1.000
2	70.21	-241.50	1.000	2	4.67	144.90	677.1	1.000
3	66.91	-241.50	1.000	3	8.47	86.08	729.0	1.683
4	50.18	-219.43	1.101	4	7.34	85.67	628.7	1.691
5	44.26	-196.36	1.230	5	7.17	76.62	549.5	1.891
6	43.59	196.36	1.230	6	7.32	85.67	627.5	1.691
7	50.95	219.43	1.101	7	8.47	86.08	729.3	1.683
8	65.10	241.50	1.000	8	4.64	144.90	673.5	1.000
9	73.82	241.50	1.000	9	5.76	144.90	835.9	1.000
10	81.02	241.50	1.000	10	6.99	144.90	1013.9	1.000
				11	3.02	65.86	199.1	2.200

Table 47: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1268$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	81.60	-235.76	1.024	1	6.27	144.90	908.5	1.000
2	69.66	-238.90	1.011	2	4.76	144.90	689.8	1.000
3	64.71	-241.50	1.000	3	9.27	90.05	834.7	1.609
4	47.10	-220.66	1.094	4	7.79	89.65	698.1	1.616
5	40.25	-198.50	1.217	5	7.00	84.75	593.1	1.710
6	39.57	198.50	1.217	6	7.77	89.65	696.3	1.616
7	47.57	220.66	1.094	7	9.26	90.05	834.3	1.609
8	63.83	241.50	1.000	8	4.74	144.90	686.6	1.000
9	72.12	238.90	1.011	9	6.30	144.90	913.0	1.000
10	80.14	235.76	1.024	10	8.01	144.90	1161.3	1.000
				11	4.76	73.57	350.3	1.970

Table 48: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1144$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.42	-241.50	1.000	1	2.82	144.90	408.2	1.000
2	5.49	-237.93	1.015	2	2.39	144.90	346.0	1.000
3	16.35	-234.06	1.032	3	3.47	89.36	310.3	1.622
4	3.54	-211.00	1.145	4	3.09	88.96	274.7	1.629
5	9.49	-187.34	1.289	5	2.19	86.59	189.9	1.673
6	4.44	187.34	1.289	6	2.58	88.96	229.5	1.629
7	8.62	211.00	1.145	7	3.54	89.36	316.3	1.622
8	13.84	234.06	1.032	8	2.23	144.90	324.0	1.000
9	11.35	237.93	1.015	9	3.12	144.90	452.7	1.000
10	18.56	241.50	1.000	10	4.60	144.90	666.5	1.000
				11	1.96	75.07	146.8	1.930

Table 49: Sizing Results for Section 3: 75% Span ($m_w = 0.0242$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 13 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2654$ m².

CONFIGURATION: CASE 14 ($\eta_f = 0.20$, $\eta_m = 0.50$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	82.69	-241.50	1.000	1	5.35	144.90	775.2	1.000
2	70.37	-241.50	1.000	2	4.35	144.90	630.3	1.000
3	65.96	-241.50	1.000	3	7.95	90.33	718.5	1.604
4	49.81	-209.85	1.151	4	7.01	89.79	629.4	1.614
5	44.15	-176.37	1.369	5	6.36	88.46	563.0	1.638
6	43.04	176.37	1.369	6	6.99	89.79	627.7	1.614
7	50.49	209.85	1.151	7	7.95	90.33	718.0	1.604
8	65.00	241.50	1.000	8	4.33	144.90	627.9	1.000
9	73.56	241.50	1.000	9	5.38	144.90	779.3	1.000
10	80.68	241.50	1.000	10	6.52	144.90	945.3	1.000
				11	3.05	73.43	223.9	1.973

Table 50: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1279$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	79.46	-241.50	1.000	1	5.82	144.90	843.9	1.000
2	71.51	-241.50	1.000	2	4.27	144.90	618.3	1.000
3	63.57	-241.50	1.000	3	9.03	94.04	849.2	1.541
4	47.67	-214.42	1.126	4	7.66	93.50	716.1	1.550
5	39.73	-181.46	1.331	5	6.45	96.48	622.3	1.502
6	39.73	181.46	1.331	6	7.66	93.50	716.1	1.550
7	47.67	214.42	1.126	7	9.03	94.04	849.2	1.541
8	63.57	241.50	1.000	8	4.27	144.90	618.3	1.000
9	71.51	241.50	1.000	9	5.82	144.90	843.9	1.000
10	79.46	241.50	1.000	10	7.53	144.90	1091.4	1.000
				11	5.33	81.27	433.6	1.783

Table 51: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1168$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.44	-241.50	1.000	1	2.84	144.90	411.2	1.000
2	7.00	-237.90	1.015	2	2.22	144.90	322.3	1.000
3	16.53	-233.97	1.032	3	3.23	93.37	301.4	1.552
4	2.53	-201.24	1.200	4	3.09	92.71	286.6	1.563
5	8.97	-167.39	1.443	5	2.19	105.38	230.9	1.375
6	3.14	167.39	1.443	6	2.62	92.71	243.1	1.563
7	8.34	201.24	1.200	7	3.39	93.37	316.5	1.552
8	14.48	233.97	1.032	8	2.18	144.90	315.8	1.000
9	11.96	237.90	1.015	9	3.07	144.90	444.7	1.000
10	18.79	241.50	1.000	10	4.54	144.90	657.4	1.000
				11	1.80	85.89	154.7	1.687

Table 52: Sizing Results for Section 3: 75% Span ($m_w = 0.0260$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 14 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2741}$ m².

CONFIGURATION: CASE 15 ($\eta_f = 0.20$, $\eta_m = 0.50$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	83.08	-241.50	1.000	1	5.26	144.90	762.9	1.000
2	69.76	-241.50	1.000	2	4.30	144.90	623.5	1.000
3	66.89	-241.50	1.000	3	7.50	91.96	689.5	1.576
4	48.99	-199.32	1.212	4	6.67	91.29	608.6	1.587
5	43.43	-154.09	1.567	5	5.69	97.17	553.3	1.491
6	42.25	154.09	1.567	6	6.65	91.29	607.1	1.587
7	49.88	199.32	1.212	7	7.50	91.96	689.4	1.576
8	65.42	241.50	1.000	8	4.28	144.90	620.5	1.000
9	73.51	241.50	1.000	9	5.29	144.90	767.5	1.000
10	80.82	241.50	1.000	10	6.41	144.90	928.9	1.000
				11	2.56	77.93	199.6	1.859

Table 53: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1299$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	77.34	-241.50	1.000	1	5.76	144.90	834.3	1.000
2	69.61	-241.50	1.000	2	4.19	144.90	607.3	1.000
3	61.87	-241.50	1.000	3	8.63	95.32	822.3	1.520
4	46.41	-212.44	1.137	4	7.34	94.65	694.9	1.531
5	38.67	-165.65	1.458	5	5.85	104.67	612.2	1.384
6	38.67	165.65	1.458	6	7.34	94.65	694.9	1.531
7	46.41	212.44	1.137	7	8.63	95.32	822.3	1.520
8	61.87	241.50	1.000	8	4.19	144.90	607.3	1.000
9	69.61	241.50	1.000	9	5.76	144.90	834.3	1.000
10	77.34	241.50	1.000	10	7.47	144.90	1083.3	1.000
				11	4.89	85.61	418.8	1.693

Table 54: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1182$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.30	-241.50	1.000	1	2.72	144.90	393.7	1.000
2	8.62	-234.17	1.031	2	2.08	144.90	302.2	1.000
3	16.35	-226.23	1.068	3	3.12	90.51	281.9	1.601
4	3.45	-186.60	1.294	4	2.81	89.93	252.8	1.611
5	9.33	-145.50	1.660	5	1.91	100.30	191.3	1.445
6	2.76	145.50	1.660	6	2.33	89.93	209.5	1.611
7	9.70	186.60	1.294	7	3.22	90.51	291.5	1.601
8	14.73	226.23	1.068	8	2.04	144.90	295.1	1.000
9	12.38	234.17	1.031	9	2.94	144.90	426.4	1.000
10	19.31	241.50	1.000	10	4.40	144.90	637.8	1.000
				11	1.77	83.06	147.3	1.745

Table 55: Sizing Results for Section 3: 75% Span ($m_w = 0.0256$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 15 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2730}$ m².

CONFIGURATION: CASE 16 ($\eta_f = 0.20$, $\eta_m = 0.53$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	86.70	-241.50	1.000	1	5.72	144.90	829.0	1.000
2	71.46	-238.50	1.013	2	4.64	144.90	673.1	1.000
3	68.26	-234.66	1.029	3	8.86	80.15	710.1	1.808
4	51.52	-215.99	1.118	4	7.62	79.83	608.4	1.815
5	45.21	-196.64	1.228	5	7.85	67.13	527.1	2.158
6	44.49	196.64	1.228	6	7.60	79.83	607.1	1.815
7	52.09	215.99	1.118	7	8.86	80.15	710.0	1.808
8	66.15	234.66	1.029	8	4.61	144.90	668.5	1.000
9	76.08	238.50	1.013	9	5.75	144.90	834.4	1.000
10	84.24	241.50	1.000	10	7.04	144.90	1020.5	1.000
				11	3.10	59.16	183.5	2.449

Table 56: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1285$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	80.59	-241.50	1.000	1	6.18	144.90	895.5	1.000
2	71.12	-241.50	1.000	2	4.62	144.90	669.4	1.000
3	64.12	-241.50	1.000	3	9.73	84.36	820.9	1.718
4	47.95	-222.37	1.086	4	8.10	84.04	680.5	1.724
5	40.26	-203.30	1.188	5	7.62	75.11	572.7	1.929
6	40.07	203.30	1.188	6	8.09	84.04	679.9	1.724
7	48.00	222.37	1.086	7	9.73	84.36	820.5	1.718
8	63.84	241.50	1.000	8	4.61	144.90	668.1	1.000
9	72.06	241.50	1.000	9	6.19	144.90	897.2	1.000
10	80.06	241.50	1.000	10	7.94	144.90	1151.8	1.000
				11	5.32	66.65	354.6	2.174

Table 57: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1141$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.17	-241.50	1.000	1	2.68	144.90	389.0	1.000
2	6.69	-234.22	1.031	2	2.17	144.90	314.3	1.000
3	15.74	-226.40	1.067	3	3.56	83.93	298.7	1.726
4	4.05	-207.07	1.166	4	3.09	83.61	258.6	1.733
5	10.55	-187.35	1.289	5	2.13	77.11	164.3	1.879
6	7.01	187.35	1.289	6	2.72	83.61	227.0	1.733
7	9.04	207.07	1.166	7	3.77	83.93	316.3	1.726
8	12.88	226.40	1.067	8	2.08	144.90	301.1	1.000
9	9.62	234.22	1.031	9	2.82	144.90	408.6	1.000
10	20.48	241.50	1.000	10	4.45	144.90	644.6	1.000
				11	2.26	68.35	154.4	2.120

Table 58: Sizing Results for Section 3: 75% Span ($m_w = 0.0241$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 16 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2667$ m².

CONFIGURATION: CASE 17 ($\eta_f = 0.20$, $\eta_m = 0.53$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	84.50	-241.50	1.000	1	5.36	144.90	777.8	1.000
2	72.26	-238.44	1.013	2	4.35	144.90	630.3	1.000
3	69.49	-234.52	1.030	3	8.09	86.06	696.6	1.684
4	50.10	-206.26	1.171	4	7.10	85.60	608.1	1.693
5	45.45	-176.57	1.368	5	6.64	81.23	539.4	1.784
6	44.02	176.57	1.368	6	7.08	85.60	605.9	1.693
7	51.70	206.26	1.171	7	8.10	86.06	697.2	1.684
8	67.39	234.52	1.030	8	4.32	144.90	626.7	1.000
9	76.25	238.44	1.013	9	5.40	144.90	782.4	1.000
10	82.28	241.50	1.000	10	6.57	144.90	952.5	1.000
				11	2.99	68.86	205.8	2.104

Table 59: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1288$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	79.62	-241.50	1.000	1	5.86	144.90	850.2	1.000
2	71.66	-241.50	1.000	2	4.28	144.90	621.0	1.000
3	63.70	-241.50	1.000	3	9.07	90.04	816.6	1.609
4	47.77	-215.99	1.118	4	7.61	89.59	682.0	1.617
5	39.81	-185.98	1.299	5	6.54	89.46	585.5	1.620
6	39.81	185.98	1.299	6	7.61	89.59	682.0	1.617
7	47.77	215.99	1.118	7	9.07	90.04	816.6	1.609
8	63.70	241.50	1.000	8	4.28	144.90	621.0	1.000
9	71.66	241.50	1.000	9	5.86	144.90	850.2	1.000
10	79.62	241.50	1.000	10	7.62	144.90	1104.8	1.000
				11	5.19	76.74	398.4	1.888

Table 60: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1156$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.79	-241.50	1.000	1	3.25	144.90	471.6	1.000
2	8.77	-236.20	1.022	2	2.60	144.90	376.2	1.000
3	15.66	-230.51	1.048	3	4.05	96.98	392.6	1.494
4	2.15	-185.95	1.299	4	3.89	96.25	374.1	1.505
5	6.25	-139.29	1.734	5	3.02	110.73	334.0	1.309
6	1.79	139.29	1.734	6	3.59	96.25	345.5	1.505
7	6.04	185.95	1.299	7	4.10	96.98	397.2	1.494
8	13.96	230.51	1.048	8	2.50	144.90	362.8	1.000
9	14.38	236.20	1.022	9	3.58	144.90	519.3	1.000
10	17.51	241.50	1.000	10	4.93	144.90	714.2	1.000
				11	2.09	87.43	182.6	1.657

Table 61: Sizing Results for Section 3: 75% Span ($m_w = 0.0277$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 17 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2721}$ m².

CONFIGURATION: CASE 18 ($\eta_f = 0.20$, $\eta_m = 0.53$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	85.76	-241.50	1.000	1	5.16	144.90	748.4	1.000
2	71.77	-238.41	1.013	2	4.18	144.90	606.2	1.000
3	68.63	-234.40	1.030	3	7.86	88.64	696.3	1.635
4	50.65	-195.57	1.235	4	6.97	88.05	614.0	1.646
5	44.64	-154.21	1.566	5	6.08	91.50	556.8	1.584
6	43.48	154.21	1.566	6	6.96	88.05	612.5	1.646
7	51.43	195.57	1.235	7	7.85	88.64	696.1	1.635
8	67.17	234.40	1.030	8	4.16	144.90	603.2	1.000
9	75.56	238.41	1.013	9	5.19	144.90	752.9	1.000
10	83.54	241.50	1.000	10	6.35	144.90	920.7	1.000
				11	2.99	74.74	223.8	1.939

Table 62: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1320$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	77.49	-241.50	1.000	1	5.94	144.90	861.8	1.000
2	69.74	-241.50	1.000	2	4.35	144.90	631.1	1.000
3	61.99	-241.50	1.000	3	8.18	92.28	754.5	1.570
4	46.50	-213.85	1.129	4	6.82	91.69	625.8	1.580
5	38.75	-169.90	1.421	5	5.44	99.40	540.6	1.458
6	38.75	169.90	1.421	6	6.82	91.69	625.8	1.580
7	46.50	213.85	1.129	7	8.18	92.28	754.5	1.570
8	61.99	241.50	1.000	8	4.35	144.90	631.1	1.000
9	69.74	241.50	1.000	9	5.94	144.90	861.8	1.000
10	77.49	241.50	1.000	10	7.71	144.90	1118.1	1.000
				11	3.97	82.56	327.4	1.755

Table 63: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1155$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.99	-241.50	1.000	1	3.02	144.90	438.4	1.000
2	9.33	-229.48	1.052	2	2.34	144.90	339.7	1.000
3	15.89	-216.68	1.115	3	4.02	90.64	364.3	1.599
4	2.41	-178.69	1.351	4	3.82	90.06	344.4	1.609
5	9.80	-139.30	1.734	5	2.80	100.45	281.5	1.443
6	3.17	139.30	1.734	6	3.35	90.06	301.9	1.609
7	8.35	178.69	1.351	7	4.09	90.64	370.6	1.599
8	14.69	216.68	1.115	8	2.30	144.90	334.1	1.000
9	13.39	229.48	1.052	9	3.28	144.90	475.7	1.000
10	18.64	241.50	1.000	10	4.71	144.90	683.2	1.000
				11	2.23	82.23	183.2	1.762

Table 64: Sizing Results for Section 3: 75% Span ($m_w = 0.0274$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 18 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2749$ m².

CONFIGURATION: CASE 19 ($\eta_f = 0.25$, $\eta_m = 0.47$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	86.55	-241.50	1.000	1	6.72	144.90	974.8	1.000
2	70.67	-239.04	1.010	2	5.64	144.90	817.6	1.000
3	69.04	-236.15	1.023	3	10.14	93.07	944.1	1.557
4	50.65	-212.32	1.137	4	9.12	92.59	844.1	1.565
5	45.51	-187.18	1.290	5	8.80	86.88	764.8	1.668
6	43.97	187.18	1.290	6	9.09	92.59	841.4	1.565
7	52.00	212.32	1.137	7	10.14	93.07	944.1	1.557
8	67.21	236.15	1.023	8	5.61	144.90	813.6	1.000
9	75.42	239.04	1.010	9	6.77	144.90	981.3	1.000
10	83.64	241.50	1.000	10	8.06	144.90	1169.3	1.000
				11	3.87	71.84	278.2	2.017

Table 65: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1387$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	80.32	-241.50	1.000	1	7.89	144.90	1143.4	1.000
2	69.30	-241.50	1.000	2	6.33	144.90	917.0	1.000
3	63.97	-241.50	1.000	3	10.19	95.85	977.0	1.512
4	47.32	-217.32	1.111	4	8.78	95.38	837.7	1.519
5	39.69	-192.80	1.253	5	7.80	94.06	734.0	1.541
6	39.38	192.80	1.253	6	8.77	95.38	836.8	1.519
7	47.66	217.32	1.111	7	10.19	95.85	977.2	1.512
8	63.08	241.50	1.000	8	6.31	144.90	914.3	1.000
9	71.18	241.50	1.000	9	7.91	144.90	1146.8	1.000
10	79.28	241.50	1.000	10	9.70	144.90	1406.3	1.000
				11	3.40	78.78	268.1	1.839

Table 66: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1195$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.71	-241.50	1.000	1	3.34	144.90	484.7	1.000
2	7.87	-236.19	1.022	2	2.72	144.90	394.4	1.000
3	14.61	-230.59	1.047	3	4.34	94.98	412.5	1.526
4	2.61	-205.36	1.176	4	4.09	94.50	386.5	1.533
5	8.03	-179.36	1.346	5	3.31	95.63	316.5	1.515
6	2.09	179.36	1.346	6	3.54	94.50	334.7	1.533
7	8.72	205.36	1.176	7	4.44	94.98	421.7	1.526
8	13.37	230.59	1.047	8	2.69	144.90	389.7	1.000
9	12.01	236.19	1.022	9	3.64	144.90	527.6	1.000
10	18.06	241.50	1.000	10	5.10	144.90	739.6	1.000
				11	2.27	80.00	181.8	1.811

Table 67: Sizing Results for Section 3: 75% Span ($m_w = 0.0257$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 19 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2838$ m².

CONFIGURATION: CASE 20 ($\eta_f = 0.25$, $\eta_m = 0.47$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	85.22	-241.50	1.000	1	6.38	144.90	924.4	1.000
2	71.58	-239.01	1.010	2	5.31	144.90	769.4	1.000
3	68.25	-236.07	1.023	3	9.20	96.03	883.1	1.509
4	50.53	-203.21	1.188	4	8.28	95.43	790.1	1.518
5	45.41	-168.07	1.437	5	7.45	96.82	720.9	1.497
6	43.73	168.07	1.437	6	8.25	95.43	787.5	1.518
7	52.03	203.21	1.188	7	9.20	96.03	883.3	1.509
8	66.93	236.07	1.023	8	5.29	144.90	766.8	1.000
9	75.27	239.01	1.010	9	6.41	144.90	929.8	1.000
10	82.77	241.50	1.000	10	7.66	144.90	1110.9	1.000
				11	3.35	77.56	259.6	1.868

Table 68: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1384$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	80.12	-241.50	1.000	1	7.71	144.90	1117.3	1.000
2	71.37	-241.50	1.000	2	6.11	144.90	886.1	1.000
3	64.07	-241.50	1.000	3	9.26	98.50	912.0	1.471
4	47.71	-208.36	1.159	4	7.95	97.90	778.4	1.480
5	40.07	-173.71	1.390	5	6.61	103.55	684.9	1.399
6	39.88	173.71	1.390	6	7.95	97.90	778.0	1.480
7	47.91	208.36	1.159	7	9.26	98.50	912.1	1.471
8	63.86	241.50	1.000	8	6.11	144.90	885.5	1.000
9	71.83	241.50	1.000	9	7.71	144.90	1118.2	1.000
10	79.83	241.50	1.000	10	9.50	144.90	1377.3	1.000
				11	2.76	84.40	232.6	1.717

Table 69: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1212$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.27	-241.50	1.000	1	3.14	144.90	454.8	1.000
2	7.43	-236.19	1.022	2	2.59	144.90	374.7	1.000
3	15.20	-230.54	1.048	3	4.17	93.21	388.3	1.554
4	2.85	-196.07	1.232	4	3.93	92.69	361.6	1.563
5	7.18	-160.25	1.507	5	3.12	104.58	326.3	1.386
6	2.20	160.25	1.507	6	3.55	92.69	342.4	1.563
7	7.36	196.07	1.232	7	4.21	93.21	408.2	1.554
8	14.18	230.54	1.048	8	2.54	144.90	367.9	1.000
9	11.73	236.19	1.022	9	3.44	144.90	499.0	1.000
10	18.26	241.50	1.000	10	4.83	144.90	700.3	1.000
				11	2.23	85.04	189.4	1.704

Table 70: Sizing Results for Section 3: 75% Span ($m_w = 0.0263$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 20 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2848$ m².

CONFIGURATION: CASE 21 ($\eta_f = 0.25$, $\eta_m = 0.47$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	84.45	-241.50	1.000	1	6.09	144.90	883.1	1.000
2	72.80	-239.01	1.010	2	5.03	144.90	729.2	1.000
3	68.33	-236.00	1.023	3	8.87	96.86	859.6	1.496
4	50.18	-193.19	1.250	4	8.05	96.13	773.9	1.507
5	44.75	-146.79	1.645	5	6.88	104.10	715.8	1.392
6	43.33	146.79	1.645	6	8.03	96.13	772.1	1.507
7	51.56	193.19	1.250	7	8.88	96.86	860.1	1.496
8	67.20	236.00	1.023	8	5.02	144.90	727.4	1.000
9	75.48	239.01	1.010	9	6.12	144.90	886.9	1.000
10	82.67	241.50	1.000	10	7.34	144.90	1063.5	1.000
				11	3.38	80.78	273.0	1.794

Table 71: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1416$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	78.88	-241.50	1.000	1	7.40	144.90	1071.9	1.000
2	71.00	-241.50	1.000	2	5.84	144.90	846.6	1.000
3	63.11	-241.50	1.000	3	8.97	98.95	887.7	1.464
4	47.33	-203.51	1.187	4	7.78	98.23	764.1	1.475
5	39.44	-156.12	1.547	5	6.22	110.17	685.1	1.315
6	39.44	156.12	1.547	6	7.78	98.23	764.1	1.475
7	47.33	203.51	1.187	7	8.97	98.95	887.7	1.464
8	63.11	241.50	1.000	8	5.84	144.90	846.6	1.000
9	71.00	241.50	1.000	9	7.40	144.90	1071.9	1.000
10	78.88	241.50	1.000	10	9.13	144.90	1322.8	1.000
				11	2.75	87.34	240.5	1.659

Table 72: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1236$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.79	-241.50	1.000	1	3.25	144.90	471.6	1.000
2	8.77	-236.20	1.022	2	2.60	144.90	376.2	1.000
3	15.66	-230.51	1.048	3	4.05	96.98	392.6	1.494
4	2.15	-185.95	1.299	4	3.89	96.25	374.1	1.505
5	6.25	-139.29	1.734	5	3.02	110.73	334.0	1.309
6	1.79	139.29	1.734	6	3.59	96.25	345.5	1.505
7	6.04	185.95	1.299	7	4.10	96.98	397.2	1.494
8	13.96	230.51	1.048	8	2.50	144.90	362.8	1.000
9	14.38	236.20	1.022	9	3.58	144.90	519.3	1.000
10	17.51	241.50	1.000	10	4.93	144.90	714.2	1.000
				11	2.09	87.43	182.6	1.657

Table 73: Sizing Results for Section 3: 75% Span ($m_w = 0.0276$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 21 yields a global weight proxy of $\mathbf{m_{w,total} = 0.2909$ m².

CONFIGURATION: CASE 22 ($\eta_f = 0.25$, $\eta_m = 0.50$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	88.79	-241.50	1.000	1	6.94	144.90	1006.5	1.000
2	73.39	-236.25	1.022	2	5.83	144.90	844.9	1.000
3	70.89	-230.28	1.049	3	9.86	87.35	861.6	1.659
4	50.58	-209.21	1.154	4	8.77	86.95	763.0	1.666
5	45.43	-187.19	1.290	5	8.78	77.91	683.7	1.860
6	43.55	187.19	1.290	6	8.74	86.95	759.7	1.666
7	52.38	209.21	1.154	7	9.87	87.35	861.8	1.659
8	68.86	230.28	1.049	8	5.80	144.90	840.8	1.000
9	78.10	236.25	1.022	9	6.98	144.90	1012.7	1.000
10	86.02	241.50	1.000	10	8.32	144.90	1206.3	1.000
				11	2.55	66.17	168.8	2.190

Table 74: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1369$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	81.06	-241.50	1.000	1	8.11	144.90	1176.2	1.000
2	71.52	-238.54	1.012	2	6.52	144.90	944.8	1.000
3	65.48	-234.69	1.029	3	9.34	90.58	845.7	1.600
4	48.06	-214.37	1.127	4	7.83	90.18	706.0	1.607
5	40.73	-192.81	1.253	5	7.02	85.38	599.6	1.697
6	40.20	192.81	1.253	6	7.81	90.18	704.7	1.607
7	48.39	214.37	1.127	7	9.33	90.58	845.3	1.600
8	64.77	234.69	1.029	8	6.50	144.90	942.1	1.000
9	73.55	238.54	1.012	9	8.14	144.90	1180.1	1.000
10	79.86	241.50	1.000	10	9.94	144.90	1441.6	1.000
				11	1.49	73.23	109.3	1.979

Table 75: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1167$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.25	-241.50	1.000	1	3.29	144.90	476.8	1.000
2	7.47	-233.06	1.036	2	2.71	144.90	392.2	1.000
3	15.13	-224.15	1.077	3	4.33	89.95	389.3	1.611
4	3.30	-202.04	1.195	4	3.99	89.56	356.9	1.618
5	8.02	-179.36	1.346	5	3.29	87.19	287.0	1.662
6	4.84	179.36	1.346	6	3.68	89.56	329.2	1.618
7	6.70	202.04	1.195	7	4.39	89.95	394.9	1.611
8	13.20	224.15	1.077	8	2.60	144.90	376.8	1.000
9	11.51	233.06	1.036	9	3.50	144.90	507.2	1.000
10	19.66	241.50	1.000	10	5.09	144.90	738.0	1.000
				11	2.17	74.69	161.9	1.940

Table 76: Sizing Results for Section 3: 75% Span ($m_w = 0.0256$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 22 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = \mathbf{0.2791}$ m².

CONFIGURATION: CASE 23 ($\eta_f = 0.25$, $\eta_m = 0.50$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	86.88	-241.50	1.000	1	6.39	144.90	926.5	1.000
2	73.23	-236.22	1.022	2	5.31	144.90	769.6	1.000
3	70.19	-230.19	1.049	3	9.32	91.53	852.9	1.583
4	51.86	-200.03	1.207	4	8.34	91.00	758.8	1.592
5	46.22	-168.08	1.437	5	7.67	89.74	688.3	1.615
6	44.69	168.08	1.437	6	8.31	91.00	756.4	1.592
7	53.10	200.03	1.207	7	9.32	91.53	852.8	1.583
8	68.76	230.19	1.049	8	5.29	144.90	766.5	1.000
9	77.15	236.22	1.022	9	6.43	144.90	931.8	1.000
10	84.46	241.50	1.000	10	7.70	144.90	1116.9	1.000
				11	3.13	73.53	229.9	1.971

Table 77: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1386$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	80.95	-241.50	1.000	1	7.67	144.90	1112.1	1.000
2	72.86	-241.50	1.000	2	6.05	144.90	876.9	1.000
3	64.76	-236.17	1.023	3	9.03	94.46	852.6	1.534
4	48.57	-206.70	1.168	4	7.64	93.92	717.6	1.543
5	40.48	-174.89	1.381	5	6.42	97.00	622.4	1.494
6	40.48	174.89	1.381	6	7.64	93.92	717.6	1.543
7	48.57	206.70	1.168	7	9.03	94.46	852.6	1.534
8	64.76	236.17	1.023	8	6.05	144.90	876.9	1.000
9	72.86	241.50	1.000	9	7.67	144.90	1112.1	1.000
10	80.95	241.50	1.000	10	9.50	144.90	1376.8	1.000
				11	2.25	80.64	181.4	1.797

Table 78: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1199$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.13	-241.50	1.000	1	3.04	144.90	440.7	1.000
2	7.97	-233.05	1.036	2	2.46	144.90	356.0	1.000
3	15.27	-224.09	1.078	3	4.17	93.21	388.3	1.554
4	3.04	-192.71	1.253	4	3.90	92.69	361.6	1.563
5	8.39	-160.26	1.507	5	3.06	98.28	300.3	1.474
6	3.24	160.26	1.507	6	3.50	92.69	324.0	1.563
7	8.35	192.71	1.253	7	4.26	93.21	397.4	1.554
8	13.67	224.09	1.078	8	2.41	144.90	348.7	1.000
9	11.73	233.05	1.036	9	3.27	144.90	473.4	1.000
10	19.16	241.50	1.000	10	4.72	144.90	684.6	1.000
				11	2.31	81.54	188.5	1.777

Table 79: Sizing Results for Section 3: 75% Span ($m_w = 0.0263$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 23 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2848$ m².

CONFIGURATION: CASE 24 ($\eta_f = 0.25$, $\eta_m = 0.50$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	87.66	-241.50	1.000	1	6.14	144.90	890.5	1.000
2	73.43	-236.20	1.022	2	5.08	144.90	737.0	1.000
3	69.63	-230.11	1.049	3	8.91	93.11	829.9	1.556
4	51.51	-189.94	1.271	4	8.04	92.45	743.3	1.567
5	45.62	-146.80	1.645	5	6.95	98.38	684.1	1.473
6	43.88	146.80	1.645	6	8.02	92.45	741.1	1.567
7	52.42	189.94	1.271	7	8.91	93.11	829.2	1.556
8	68.80	230.11	1.049	8	5.07	144.90	734.6	1.000
9	77.33	236.20	1.022	9	6.18	144.90	896.2	1.000
10	84.97	241.50	1.000	10	7.44	144.90	1078.0	1.000
				11	3.02	77.86	234.8	1.861

Table 80: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1415$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	78.92	-241.50	1.000	1	7.30	144.90	1057.8	1.000
2	71.03	-241.50	1.000	2	5.73	144.90	830.0	1.000
3	63.14	-241.50	1.000	3	8.88	95.65	849.8	1.515
4	47.35	-204.47	1.181	4	7.64	94.99	725.4	1.525
5	39.46	-159.39	1.515	5	6.13	105.09	644.7	1.379
6	39.46	159.39	1.515	6	7.64	94.99	725.4	1.525
7	47.35	204.47	1.181	7	8.88	95.65	849.8	1.515
8	63.14	241.50	1.000	8	5.73	144.90	830.0	1.000
9	71.03	241.50	1.000	9	7.30	144.90	1057.8	1.000
10	78.92	241.50	1.000	10	9.07	144.90	1314.2	1.000
				11	2.58	84.75	218.8	1.710

Table 81: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1219$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.99	-241.50	1.000	1	3.02	144.90	438.4	1.000
2	9.33	-229.48	1.052	2	2.34	144.90	339.7	1.000
3	15.89	-216.68	1.115	3	4.02	90.64	364.3	1.599
4	2.41	-178.69	1.351	4	3.82	90.06	344.4	1.609
5	9.80	-139.30	1.734	5	2.80	100.45	281.5	1.443
6	3.17	139.30	1.734	6	3.35	90.06	301.9	1.609
7	8.35	178.69	1.351	7	4.09	90.64	370.6	1.599
8	14.69	216.68	1.115	8	2.30	144.90	334.1	1.000
9	13.39	229.48	1.052	9	3.28	144.90	475.7	1.000
10	18.64	241.50	1.000	10	4.71	144.90	683.2	1.000
				11	2.23	82.23	183.2	1.762

Table 82: Sizing Results for Section 3: 75% Span ($m_w = 0.0276$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 24 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2891$ m².

CONFIGURATION: CASE 25 ($\eta_f = 0.25$, $\eta_m = 0.53$, $\eta_r = 0.65$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	90.13	-241.50	1.000	1	6.88	144.90	998.2	1.000
2	74.12	-233.02	1.036	2	5.77	144.90	836.9	1.000
3	73.09	-223.44	1.081	3	10.14	80.98	821.3	1.789
4	54.44	-205.64	1.174	4	8.89	80.66	716.8	1.796
5	47.02	-187.19	1.290	5	9.34	67.96	634.6	2.132
6	46.59	187.19	1.290	6	8.88	80.66	716.0	1.796
7	55.83	205.64	1.174	7	10.17	80.98	823.2	1.789
8	70.12	223.44	1.081	8	5.74	144.90	832.6	1.000
9	77.88	233.02	1.036	9	6.91	144.90	1002.1	1.000
10	88.40	241.50	1.000	10	8.28	144.90	1201.5	1.000
				11	2.31	59.34	136.9	2.442

Table 83: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1377$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	82.64	-241.50	1.000	1	7.63	144.90	1106.2	1.000
2	74.11	-235.54	1.025	2	5.99	144.90	869.3	1.000
3	66.68	-228.21	1.058	3	10.58	84.56	894.8	1.714
4	49.54	-210.92	1.145	4	8.94	84.24	752.9	1.720
5	41.43	-192.81	1.253	5	8.55	75.40	644.4	1.922
6	41.39	192.81	1.253	6	8.94	84.24	752.8	1.720
7	49.62	210.92	1.145	7	10.58	84.56	894.9	1.714
8	66.16	228.21	1.058	8	5.98	144.90	867.8	1.000
9	75.05	235.54	1.025	9	7.64	144.90	1107.8	1.000
10	82.17	241.50	1.000	10	9.50	144.90	1377.2	1.000
				11	3.50	66.30	232.1	2.185

Table 84: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1200$ m²)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.95	-241.50	1.000	1	3.10	144.90	449.8	1.000
2	6.27	-229.50	1.052	2	2.62	144.90	379.8	1.000
3	15.17	-216.80	1.114	3	4.63	84.20	390.2	1.721
4	5.43	-198.28	1.218	4	4.03	83.88	337.9	1.727
5	9.87	-179.37	1.346	5	3.25	77.39	251.9	1.872
6	5.61	179.37	1.346	6	3.59	83.88	300.8	1.727
7	8.81	198.28	1.218	7	4.58	84.20	385.7	1.721
8	13.82	216.80	1.114	8	2.49	144.90	361.0	1.000
9	11.43	229.50	1.052	9	3.37	144.90	488.5	1.000
10	18.65	241.50	1.000	10	4.88	144.90	707.5	1.000
				11	2.51	67.98	170.3	2.132

Table 85: Sizing Results for Section 3: 75% Span ($m_w = 0.0256$ m²)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 25 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2833$ m².

CONFIGURATION: CASE 26 ($\eta_f = 0.25$, $\eta_m = 0.53$, $\eta_r = 0.70$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	90.18	-241.50	1.000	1	6.56	144.90	950.7	1.000
2	74.33	-232.98	1.037	2	5.47	144.90	793.3	1.000
3	72.67	-223.33	1.081	3	9.00	86.82	781.7	1.669
4	52.72	-196.39	1.230	4	7.96	86.37	687.6	1.678
5	47.26	-168.08	1.437	5	7.50	82.05	615.4	1.766
6	45.87	168.08	1.437	6	7.94	86.37	685.5	1.678
7	54.29	196.39	1.230	7	9.01	86.82	782.4	1.669
8	70.18	223.33	1.081	8	5.44	144.90	788.9	1.000
9	78.98	232.98	1.037	9	6.59	144.90	956.2	1.000
10	87.68	241.50	1.000	10	7.92	144.90	1148.7	1.000
				11	1.98	68.83	136.0	2.105

Table 86: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1373 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	81.05	-241.50	1.000	1	7.37	144.90	1067.8	1.000
2	72.94	-241.50	1.000	2	5.72	144.90	829.5	1.000
3	64.84	-235.15	1.027	3	9.48	90.12	854.2	1.608
4	48.63	-208.03	1.161	4	8.01	89.67	718.1	1.616
5	40.52	-179.09	1.348	5	6.92	89.65	620.4	1.616
6	40.52	179.09	1.348	6	8.01	89.67	718.1	1.616
7	48.63	208.03	1.161	7	9.48	90.12	854.2	1.608
8	64.84	235.15	1.027	8	5.72	144.90	829.5	1.000
9	72.94	241.50	1.000	9	7.37	144.90	1067.8	1.000
10	81.05	241.50	1.000	10	9.24	144.90	1339.5	1.000
				11	3.02	76.08	230.0	1.905

Table 87: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1198 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	22.37	-241.50	1.000	1	3.01	144.90	436.8	1.000
2	6.80	-229.48	1.052	2	2.52	144.90	365.5	1.000
3	16.23	-216.73	1.114	3	4.07	89.10	362.8	1.626
4	4.14	-188.90	1.278	4	3.69	88.65	327.1	1.634
5	9.60	-160.26	1.507	5	2.82	91.15	256.9	1.590
6	5.02	160.26	1.507	6	3.31	88.65	293.6	1.634
7	8.58	188.90	1.278	7	4.13	89.10	367.6	1.626
8	13.96	216.73	1.114	8	2.40	144.90	347.8	1.000
9	12.08	229.48	1.052	9	3.27	144.90	474.3	1.000
10	18.96	241.50	1.000	10	4.71	144.90	683.4	1.000
				11	2.04	77.23	157.9	1.876

Table 88: Sizing Results for Section 3: 75% Span ($m_w = 0.0261 \text{ m}^2$)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 26 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2832 \text{ m}^2$.

CONFIGURATION: CASE 27 ($\eta_f = 0.25$, $\eta_m = 0.53$, $\eta_r = 0.75$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	89.16	-241.50	1.000	1	6.20	144.90	899.2	1.000
2	75.64	-232.95	1.037	2	5.12	144.90	742.9	1.000
3	72.51	-223.24	1.082	3	8.84	89.34	790.0	1.622
4	53.09	-186.22	1.297	4	7.91	88.76	702.4	1.632
5	46.76	-146.80	1.645	5	6.95	92.28	641.5	1.570
6	45.51	146.80	1.645	6	7.89	88.76	700.7	1.632
7	54.18	186.22	1.297	7	8.84	89.34	790.2	1.622
8	70.77	223.24	1.082	8	5.10	144.90	739.7	1.000
9	79.49	232.95	1.037	9	6.23	144.90	903.9	1.000
10	86.98	241.50	1.000	10	7.52	144.90	1090.3	1.000
				11	2.56	74.53	190.6	1.944

Table 89: Sizing Results for Section 1: Fuselage Root ($m_w = 0.1413 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	79.00	-241.50	1.000	1	7.18	144.90	1041.2	1.000
2	71.10	-241.50	1.000	2	5.59	144.90	810.4	1.000
3	63.20	-241.50	1.000	3	8.84	92.27	816.0	1.570
4	47.40	-205.63	1.174	4	7.53	91.69	690.7	1.580
5	39.50	-163.33	1.479	5	6.11	99.49	607.7	1.456
6	39.50	163.33	1.479	6	7.53	91.69	690.7	1.580
7	47.40	205.63	1.174	7	8.84	92.27	816.0	1.570
8	63.20	241.50	1.000	8	5.59	144.90	810.4	1.000
9	71.10	241.50	1.000	9	7.18	144.90	1041.2	1.000
10	79.00	241.50	1.000	10	9.00	144.90	1304.5	1.000
				11	2.50	81.67	204.3	1.774

Table 90: Sizing Results for Section 2: Aerodynamic Kink ($m_w = 0.1204 \text{ m}^2$)

Boom Sizing				Panel Sizing				
ID	A (cm ²)	σ (MPa)	$\mathbf{RF}_\sigma$	ID	t (mm)	τ (MPa)	q (kN/m)	$\mathbf{RF}_\tau$
1	21.99	-241.50	1.000	1	3.02	144.90	438.4	1.000
2	9.33	-229.48	1.052	2	2.34	144.90	339.7	1.000
3	15.89	-216.68	1.115	3	4.02	90.64	364.3	1.599
4	2.41	-178.69	1.351	4	3.82	90.06	344.4	1.609
5	9.80	-139.30	1.734	5	2.80	100.45	281.5	1.443
6	3.17	139.30	1.734	6	3.35	90.06	301.9	1.609
7	8.35	178.69	1.351	7	4.09	90.64	370.6	1.599
8	14.69	216.68	1.115	8	2.30	144.90	334.1	1.000
9	13.39	229.48	1.052	9	3.28	144.90	475.7	1.000
10	18.64	241.50	1.000	10	4.71	144.90	683.2	1.000
				11	2.23	82.23	183.2	1.762

Table 91: Sizing Results for Section 3: 75% Span ($m_w = 0.0274 \text{ m}^2$)

Configuration Weight Metric: Summing the optimal cross-sectional areas from the three stations for CASE 27 yields a global weight proxy of $\mathbf{m}_{w,\text{total}} = 0.2891 \text{ m}^2$.